



BITS Pilani

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DE ZG 631 Materials Technology & Testing-Lecture 1 (Structure properties of Materials)

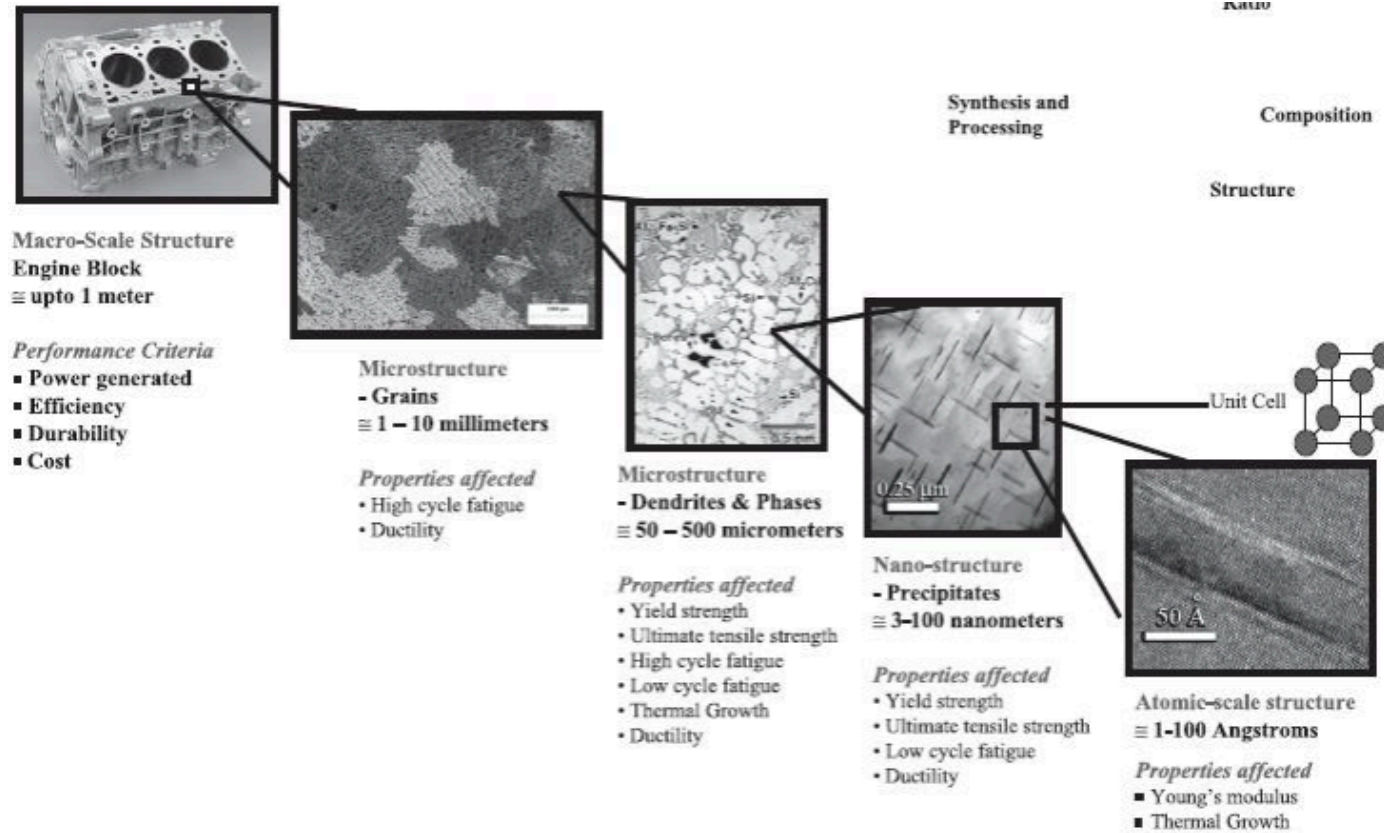
Prof.P Srinivasan

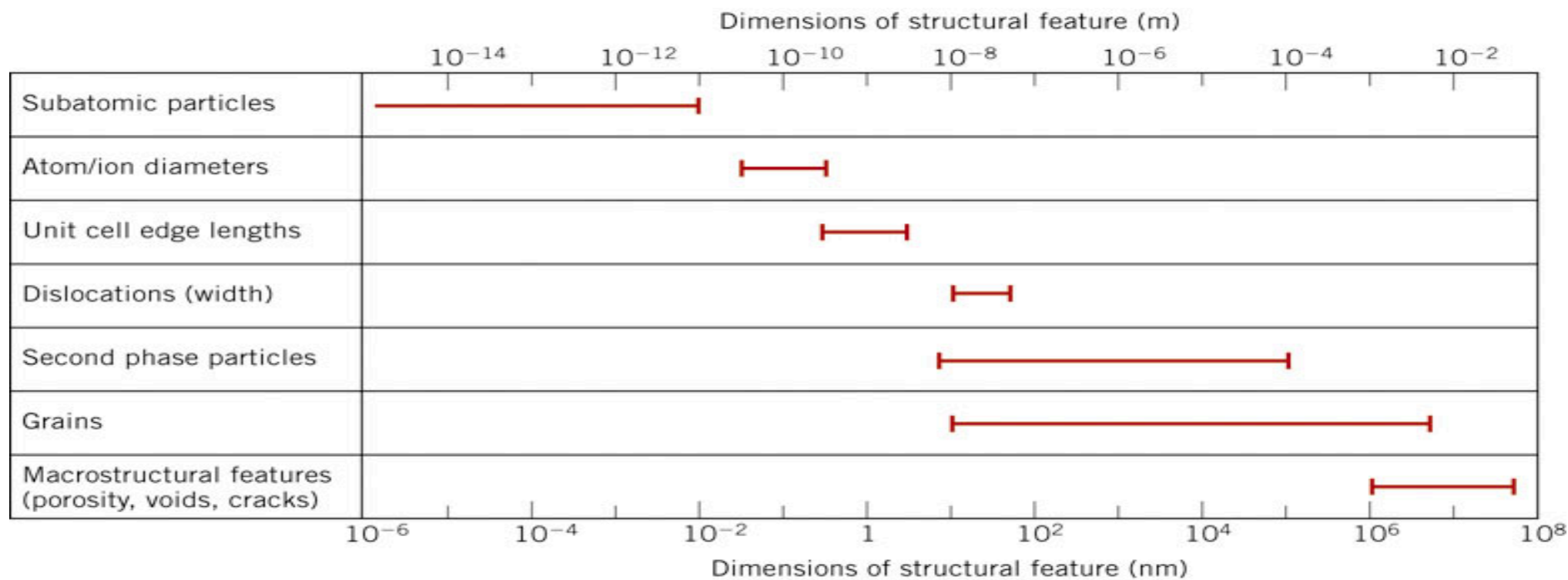
Agenda



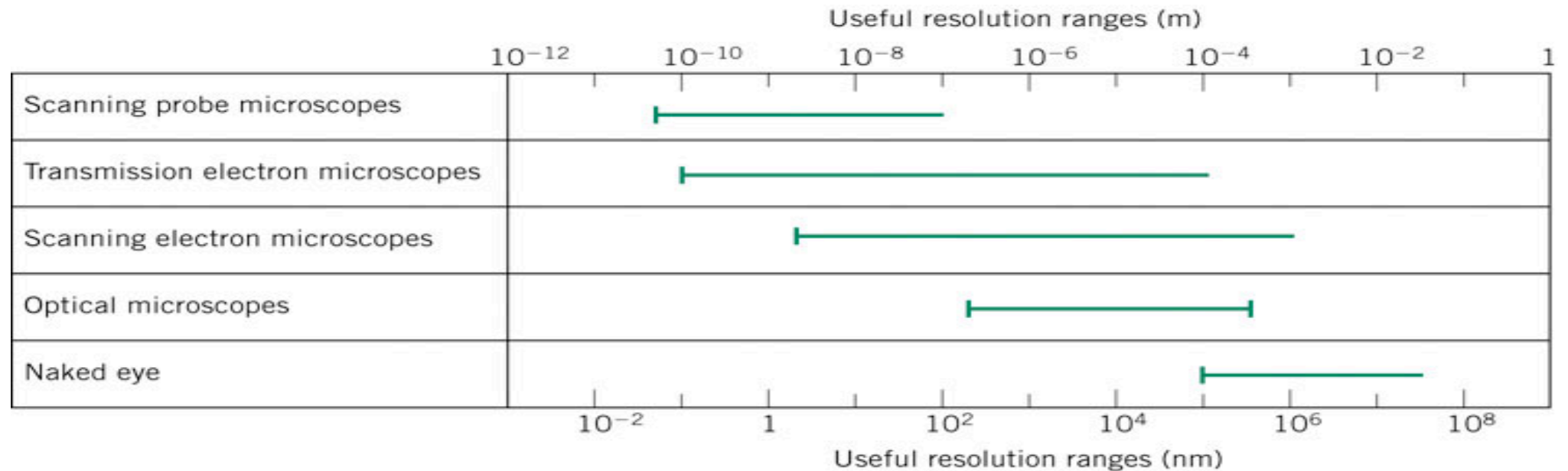
- Structure and Properties correlation
- Atomic structure and it's impact on properties
- Crystal structure and it's impact on properties
- Microstructure formation and it's impact on properties
- ASTM grain size and it's applications
- Conclusion

Stages of analysis and properties of Materials





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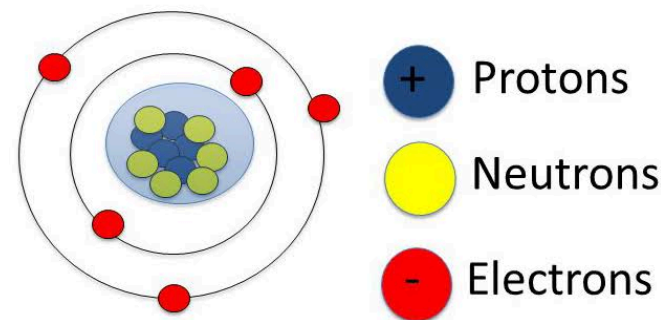


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Atomic Structure



Atom – Electrons – 9.11×10^{-31} kg
 Protons } 1.67×10^{-27} kg
 Neutrons }



Atomic Number = No. of **protons** in nucleus of atom
= No. of **electrons** of *neutral species*

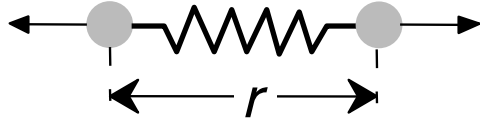
A [=] **atomic mass unit** = 1 amu = $1/12$ mass of ^{12}C (C 12.011)

Atomic wt = wt of 6.023×10^{23} molecules or atoms
1 amu/atom = 1g/mol

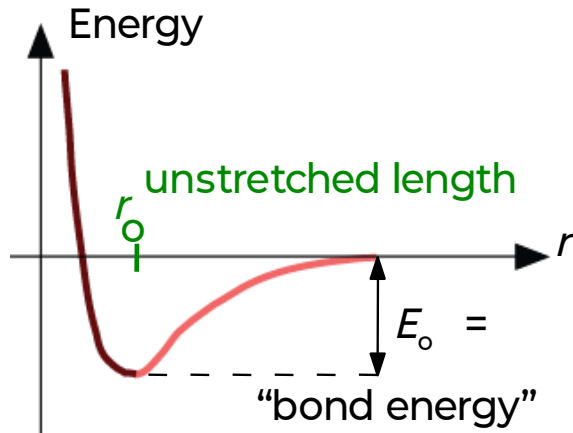
Properties From Bonding: T_m



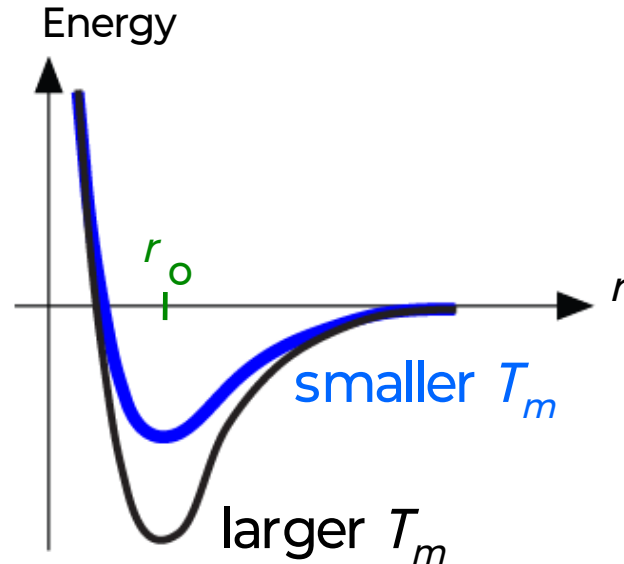
- Bond length, r



- Bond energy, E_o

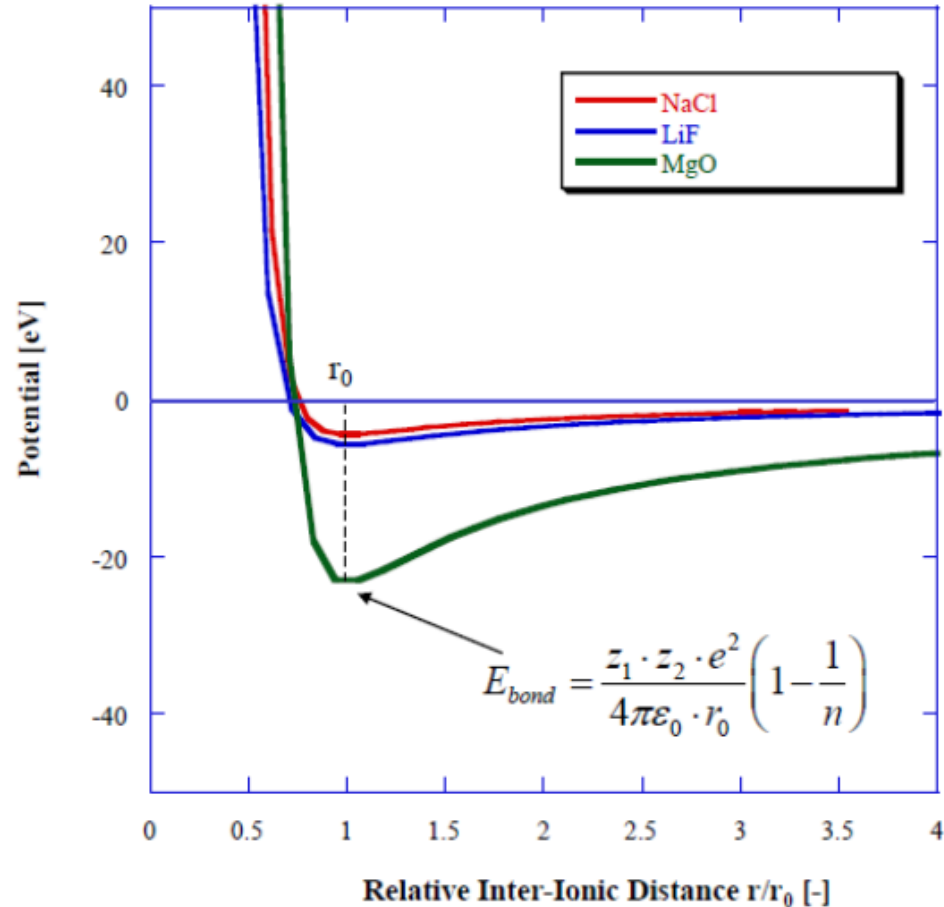


- Melting Temperature, T_m



T_m is larger if E_o is larger.

Melting Temperature: An Illustration

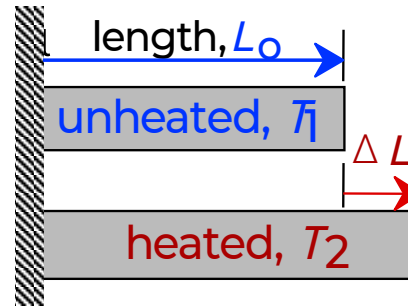


MgO	NaCl	LiF	Crystal Structure
2852°C	801°C	848°C	Rocksalt

Properties From Bonding : α



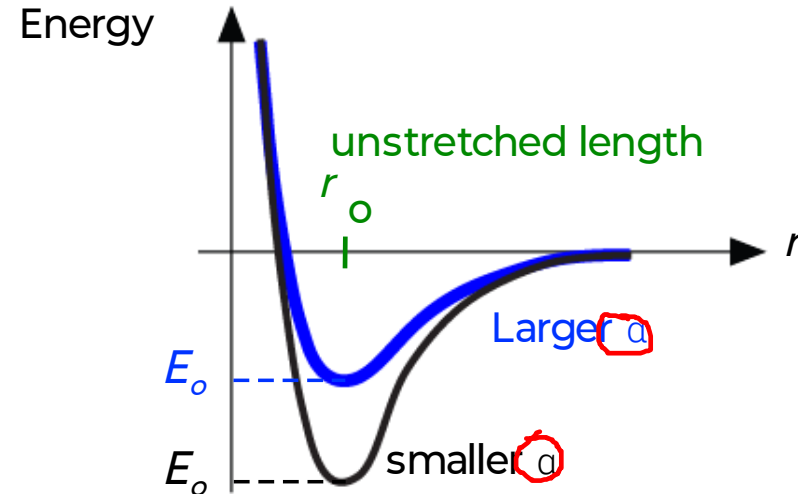
- Coefficient of thermal expansion,



coeff. thermal expansion

$$\frac{\Delta L}{L_0} = \alpha (T_2 - T_1)$$

- $\alpha \sim$ symmetry at r_0



α is larger if E_0 is smaller.

Ceramics

(Ionic & covalent bonding):

Large bond energy

large T_m

large E

small α

Metals

(Metallic bonding):

Variable bond energy

moderate T_m

moderate E

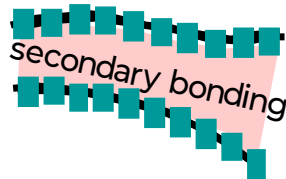
moderate α

Polymers

(Covalent & Secondary):

Directional Properties

Secondary bonding dominates



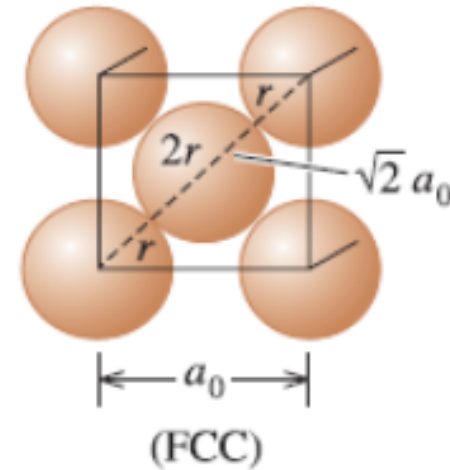
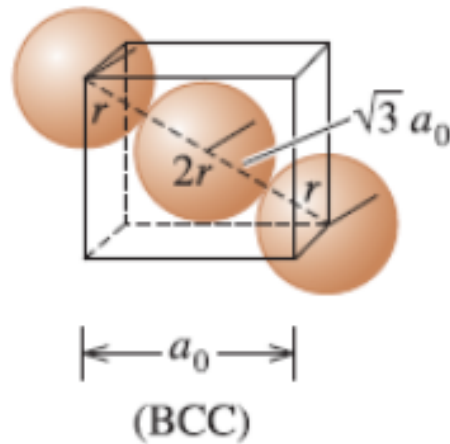
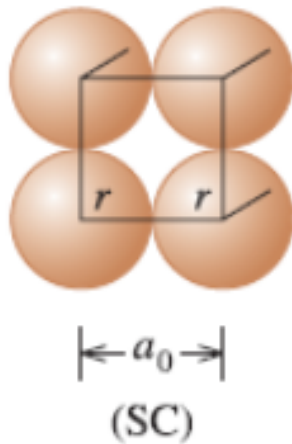
small T_m

small E

large α

Some Important Points:

Determine the relationship between the atomic radius (r) and the lattice parameter (a_0) in SC, BCC, and FCC structures when one atom is located at each lattice point.



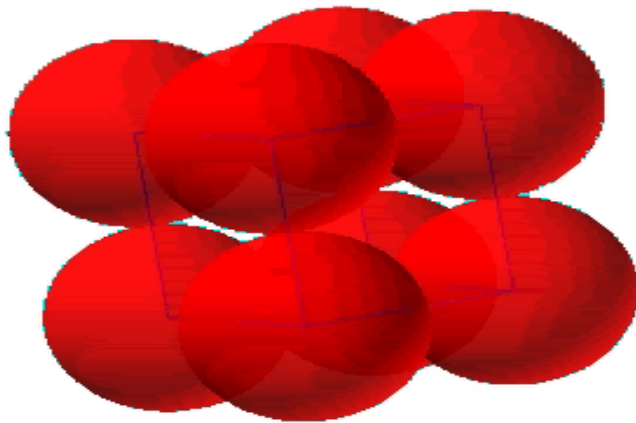
Close Packed Direction in SC is $\langle 100 \rangle$

BCC: $\langle 111 \rangle$

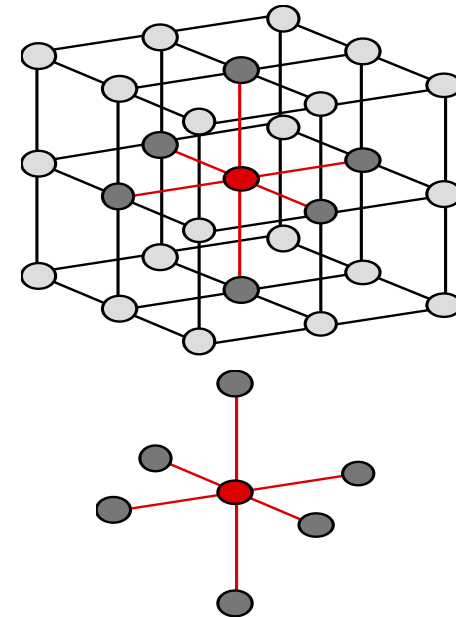
FCC: $\langle 110 \rangle$

SIMPLE CUBIC STRUCTURE (SC)

- Rare due to poor packing (only Po has this structure)
- Close-packed directions are cube edges.



- Coordination # = 6
(# nearest neighbors)



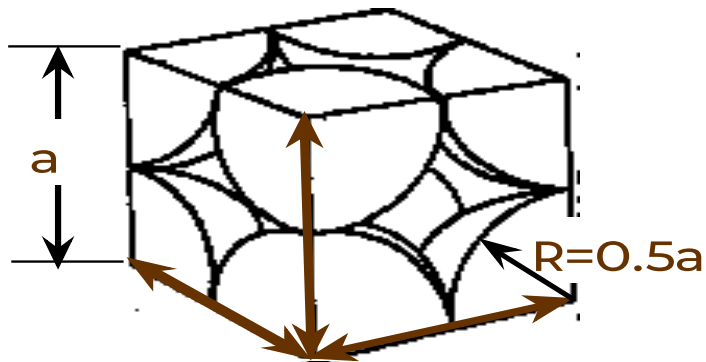
ATOMIC PACKING FACTOR



$$\text{APF} = \frac{\text{Volume of atoms in unit cell}^*}{\text{Volume of unit cell}}$$

*assume hard spheres

- APF for a simple cubic structure = 0.52



close-packed directions

contains $8 \times 1/8 =$

1 atom/unit cell

$$\text{APF} = \frac{\text{atoms unit cell} \times \frac{4}{3} \pi (0.5a)^3}{a^3}$$

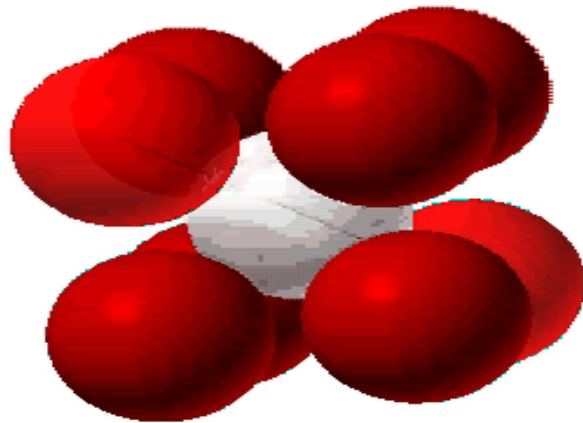
volume atom

volume unit cell

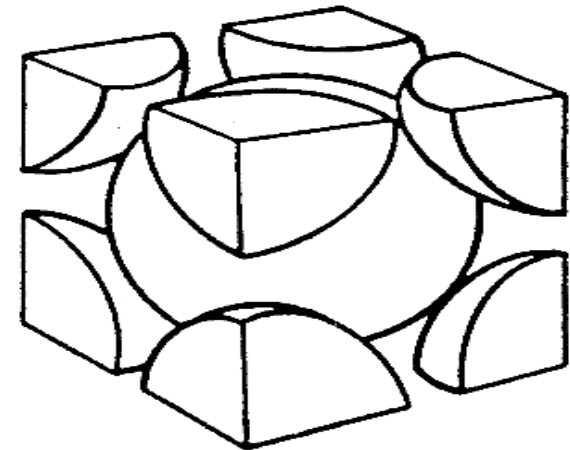
BODY CENTERED CUBIC STRUCTURE (BCC)

- Close packed directions are cube diagonals.
 - Note: All atoms are identical; the center atom is shaded differently only for ease of viewing.

Iron, chromium ,tungsten, tantalum etc.



- Coordination # = 8



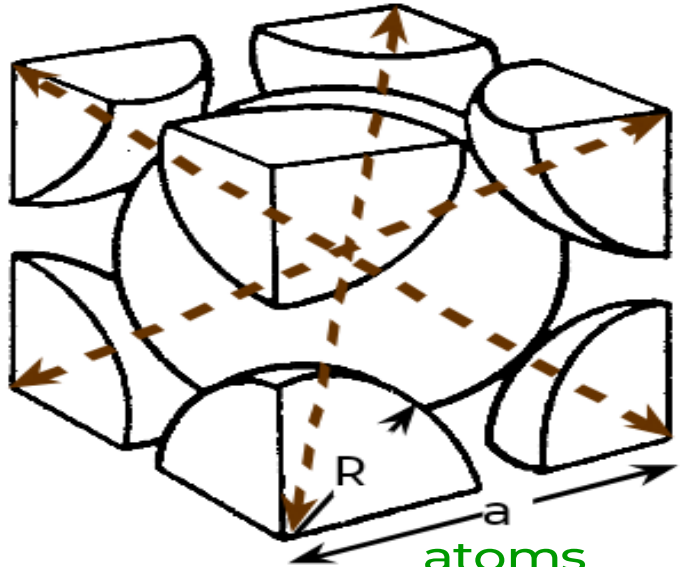
ATOMIC PACKING FACTOR: BCC

innovate

achieve

lead

- APF for a body-centered cubic structure = 0.68



Close-packed directions:

$$\begin{aligned} \text{length} &= 4R \\ &= \sqrt{3} a \end{aligned}$$

Unit cell contains:

$$\begin{aligned} &1 + 8 \times 1/8 \\ &= 2 \text{ atoms/unit cell} \end{aligned}$$

atoms
unit cell

APF =

2

$$\frac{4}{3} \pi \left(\frac{\sqrt{3} a}{4} \right)^3$$

volume
atom

a^3

volume
unit cell

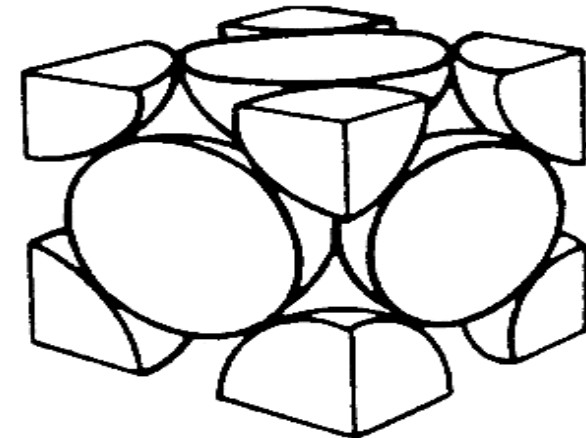
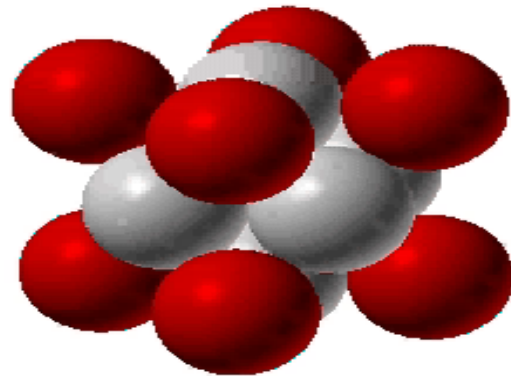
FACE CENTERED CUBIC STRUCTURE (FCC)

- Close packed directions are face diagonals.

--Note: All atoms are identical; the face-centered atoms are shaded differently only for ease of viewing.

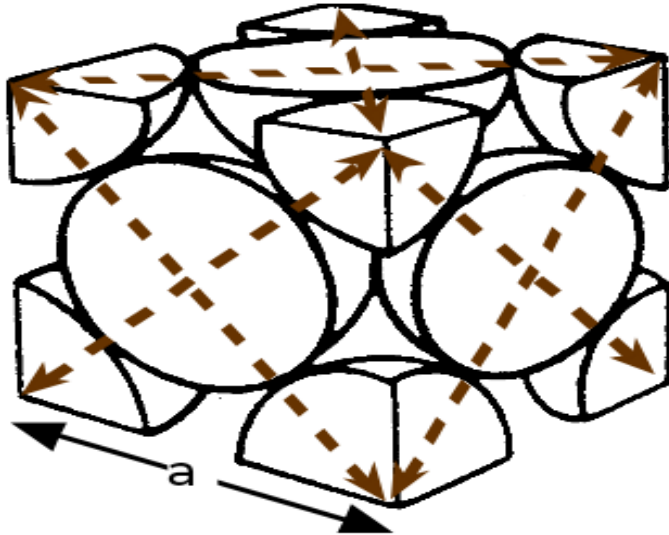
Aluminum, nickel, copper, gold, silver etc.

- Coordination # = 12



ATOMIC PACKING FACTOR: FCC

- APF for a body-centered cubic structure = 0.74



C lose-packed directions:

$$\begin{aligned} \text{length} &= 4R \\ &= \frac{\sqrt{2}}{2} a \end{aligned}$$

Unit cell contains:

$$\begin{aligned} &6 \times \frac{1}{2} + 8 \times \frac{1}{8} \\ &= 4 \text{ atoms/unit cell} \end{aligned}$$

$$\text{APF} = \frac{\frac{\text{atoms}}{\text{unit cell}} \times \frac{\text{volume}}{\text{atom}}}{\frac{\text{volume}}{\text{unit cell}}} = \frac{4 \times \frac{4}{3} \pi \left(\frac{\sqrt{2} a}{4} \right)^3}{a^3}$$

THEORETICAL DENSITY, ρ

atoms/unit cell

Atomic weight (g/mol)

$$\rho = \frac{n A}{V_c N_A}$$

Volume/unit cell
(cm³/unit cell)

$$V_c N_A$$

Avogadro's number

(6.023 × 10²³ atoms/mol)

Example: Copper

- crystal structure = FCC: 4 atoms/unit cell
- atomic weight = 63.55 g/mol (1 amu = 1 g/mol)
- atomic radius R = 0.128 nm (1 nm = 10⁻⁷ cm)

$$V_c = a^3 ; \text{ For FCC, } a = 4R/\sqrt{2}; \quad V_c = 4.75 \times 10^{-23} \text{ cm}^3$$

Result: theoretical $\rho_{\text{Cu}} = 8.89 \text{ g/cm}^3$

Compare to actual: $\rho_{\text{Cu}} = 8.94 \text{ g/cm}^3$

Characteristics of Selected Elements at 20C



Element	Symbol	At. Weight (amu)	Atomic (nm)	radius	Crystal Structure	Density (g/cm ³)
Aluminum	Al	26.98	0.143		FCC	2.71
Argon	Ar	39.95	-----		-----	-----
Barium	Ba	137.33	0.217		BCC	3.5
Beryllium	Be	9.012	0.114		HCP	1.85
Boron	B	10.81	-----		Rhomb	2.34
Bromine	Br	79.90	-----		-----	-----
Cadmium	Cd	112.41	0.149		HCP	8.65
Calcium	Ca	40.08	0.197		FCC	1.55
Carbon	C	12.011	0.071		Hex	2.25
Cesium	Cs	132.91	0.265		BCC	1.87
Chlorine	Cl	35.45	-----		-----	-----
Chromium	Cr	52.00	0.125		BCC	7.19
Cobalt	Co	58.93	0.125		HCP	8.9
Copper	Cu	63.55	0.128		FCC	8.94
Flourine	F	19.00	-----		-----	-----
Gallium	Ga	69.72	0.122		Ortho.	5.90
Germanium	Ge	72.59	0.122		Dia. cubic	5.32
Gold	Au	196.97	0.144		FCC	19.32
Helium	He	4.003	-----		-----	-----
Hydrogen	H	1.008	-----		-----	-----

DENSITIES OF MATERIAL CLASSES

Why?

Metals have...

- close-packing
(metallic bonding)
- large atomic mass

Ceramics have...

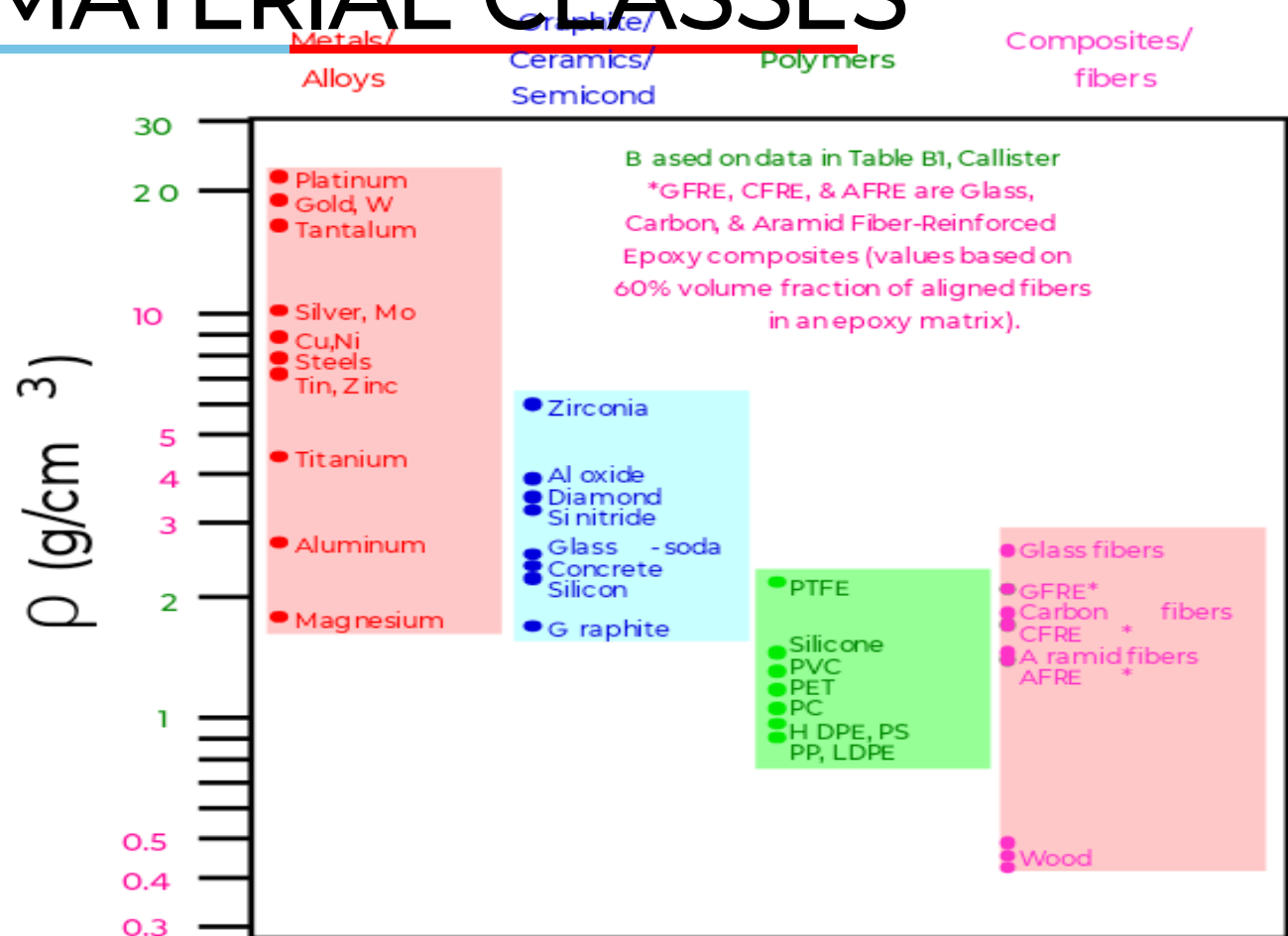
- less dense packing
(covalent bonding)
- often lighter elements

Polymers have...

- poor packing
(often amorphous)
- lighter elements (C,H,O)

Composites have...

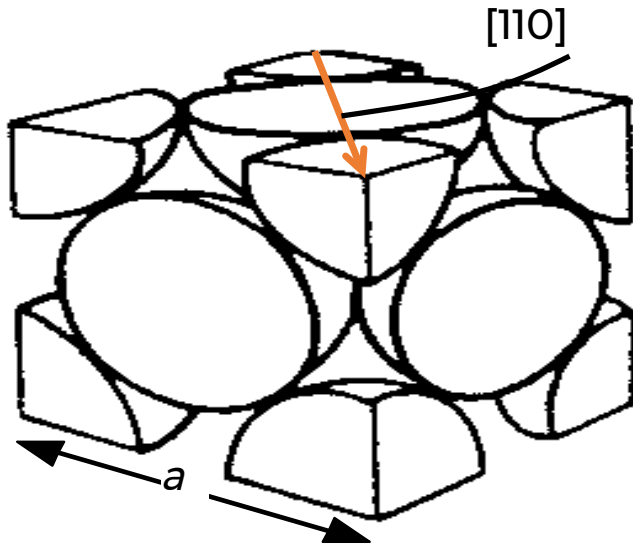
- intermediate values



Linear Density

- Linear Density of Atoms $\equiv$ LD =

$$\frac{\text{Number of atoms}}{\text{Unit length of direction vector}}$$



ex: linear density of Al in $[110]$ direction

$$a = 0.405 \text{ nm}$$

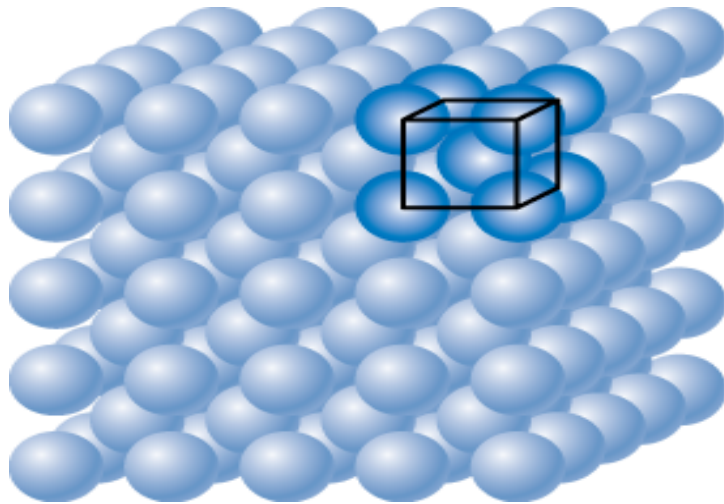
$$\text{LD} = \frac{\text{\# atoms}}{\text{length}} = \frac{2}{\sqrt{2}a} = 3.5 \text{ nm}^{-1}$$

Atomic Packing of Crystallographic Planes

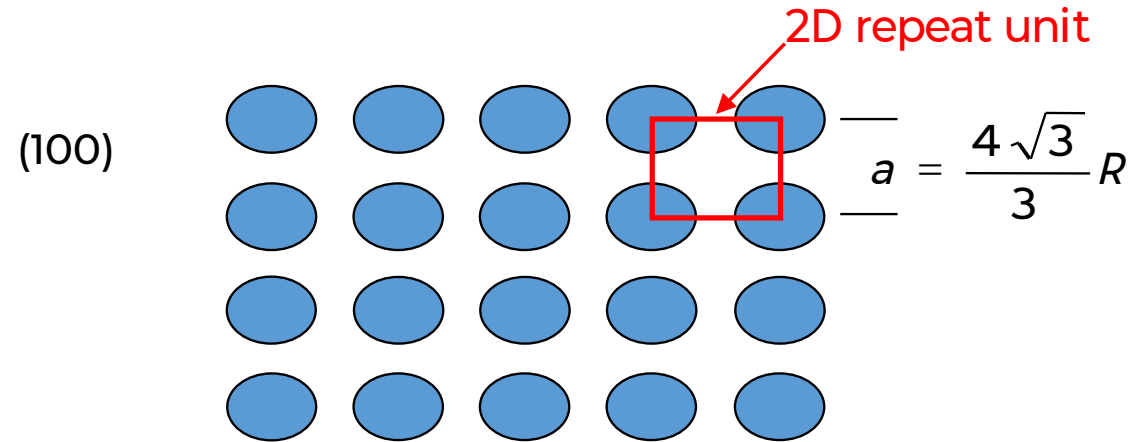
- We want to examine the atomic packing of crystallographic planes
 - a) Draw (100) and (111) crystallographic planes for Fe.
 - b) Calculate the planar density for each of these planes.

Planar Density of (100) Iron

Solution: At $T < 912^\circ\text{C}$ iron has the BCC structure.



From Fig. 4.2(c), Callister's Materials Science and Engineering, Adapted Version.



Radius of iron $R = 0.1241\text{ nm}$

$$\text{Planar Density} = \frac{\text{atoms}}{\text{area}} = \frac{1}{a^2} = \frac{1}{\left(\frac{4\sqrt{3}}{3}R\right)^2} = 12.1 \frac{\text{atoms}}{\text{nm}^2} = 1.2 \times 10^{19} \frac{\text{atoms}}{\text{m}^2}$$

atoms

2D repeat unit

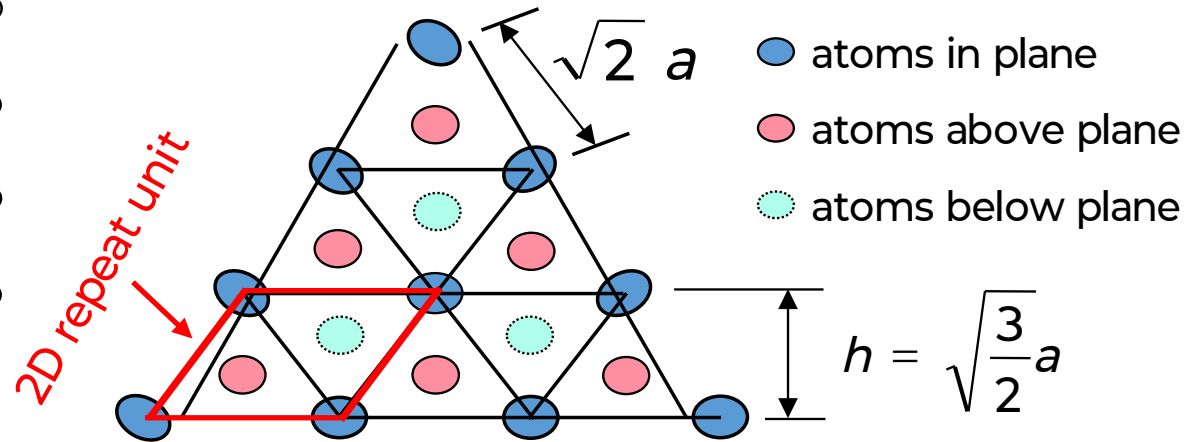
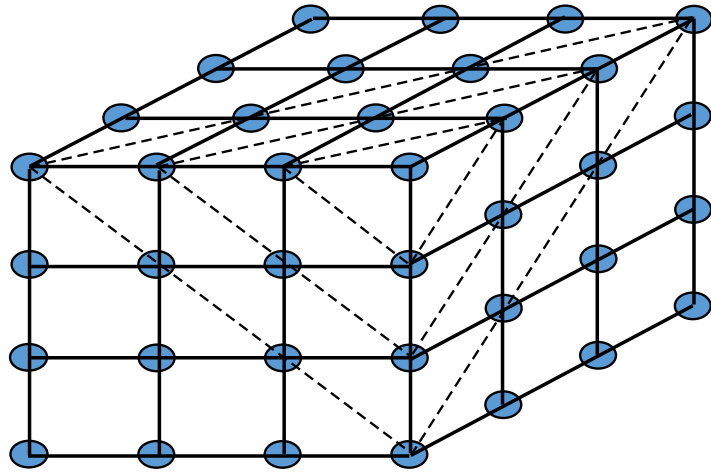
area

2D repeat unit

Planar Density of (111) Iron

Solution (cont): (111) plane

1 atom in plane/ unit surface cell



$$\text{area} = \sqrt{2} ah = \sqrt{3} a^2 = \sqrt{3} \left(\frac{4\sqrt{3}}{3} R \right)^2 = \frac{16\sqrt{3}}{3} R^2$$

atoms
2D repeat unit

Planar Density =

area
2D repeat unit

$$= \frac{1}{\frac{16\sqrt{3}}{3} R^2} = 7.0 \frac{\text{atoms}}{\text{nm}^2}$$

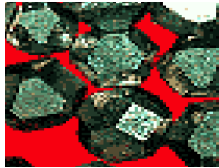
$$0.70 \times 10^{19} \frac{\text{atoms}}{\text{m}^2}$$

CRYSTALS AS BUILDING BLOCKS



- Some engineering applications require single crystals:

--diamond single
crystals for abrasives



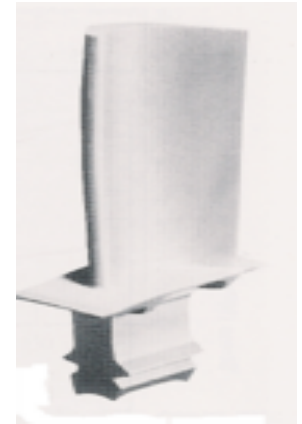
(Courtesy Martin Deakins,
GE Superabrasives, Worthington, OH.
Used with permission.)

- Crystal properties reveal features
of atomic structure.

--Ex: Certain crystal planes in quartz
fracture more easily than others.

--turbine blades

Fig. 8.30(c), *Callister 6e*.
(Fig. 8.30(c) courtesy
of Pratt and Whitney).



(Courtesy P.M. Anderson)

POLYCRYSTALS



- Most engineering materials are **polycrystals**. (Composed of many small crystals or grains)



Adapted from Fig. K, color inset pages of *Callister 6e*.
(Fig. K is courtesy of Paul E. Danielson, Teledyne Wah Chang Albany)

- Nb-Hf-W plate with an electron beam weld.
- Each "grain" is a single crystal.
- If crystals are randomly oriented,
 overall component properties are not directional.
- Crystal sizes typ. range from 1 nm to 2 cm
 (i.e., from a few to millions of atomic layers).

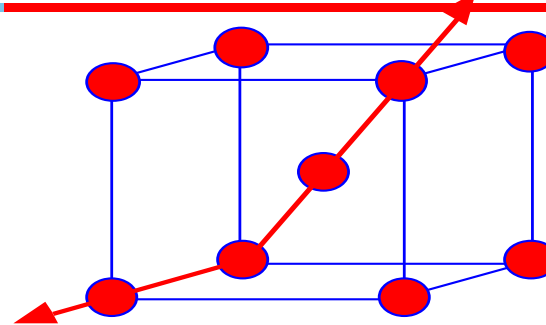
SINGLE VS POLYCRYSTALS



- Single Crystals

- Properties vary with direction: **anisotropic**.
- Example: the modulus of elasticity (E) in BCC iron:

E (diagonal) = 273 GPa



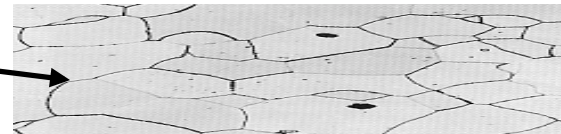
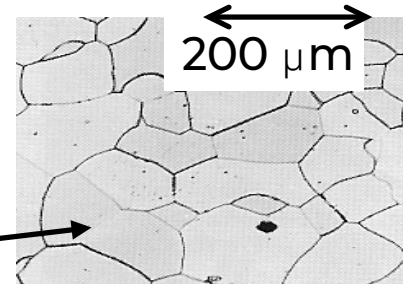
E (edge) = 125 GPa

Data from Table 3.3, *Callister 6e*.
(Source of data is R.W. Hertzberg, *Deformation and Fracture Mechanics of Engineering Materials*, 3rd ed., John Wiley and Sons, 1989.)

- Polycrystals

- Properties may/may not vary with direction.
- If grains are randomly oriented: **isotropic**.
($E_{\text{poly iron}} = 210 \text{ GPa}$)
- If grains are **textured**, anisotropic.

SUMMARY



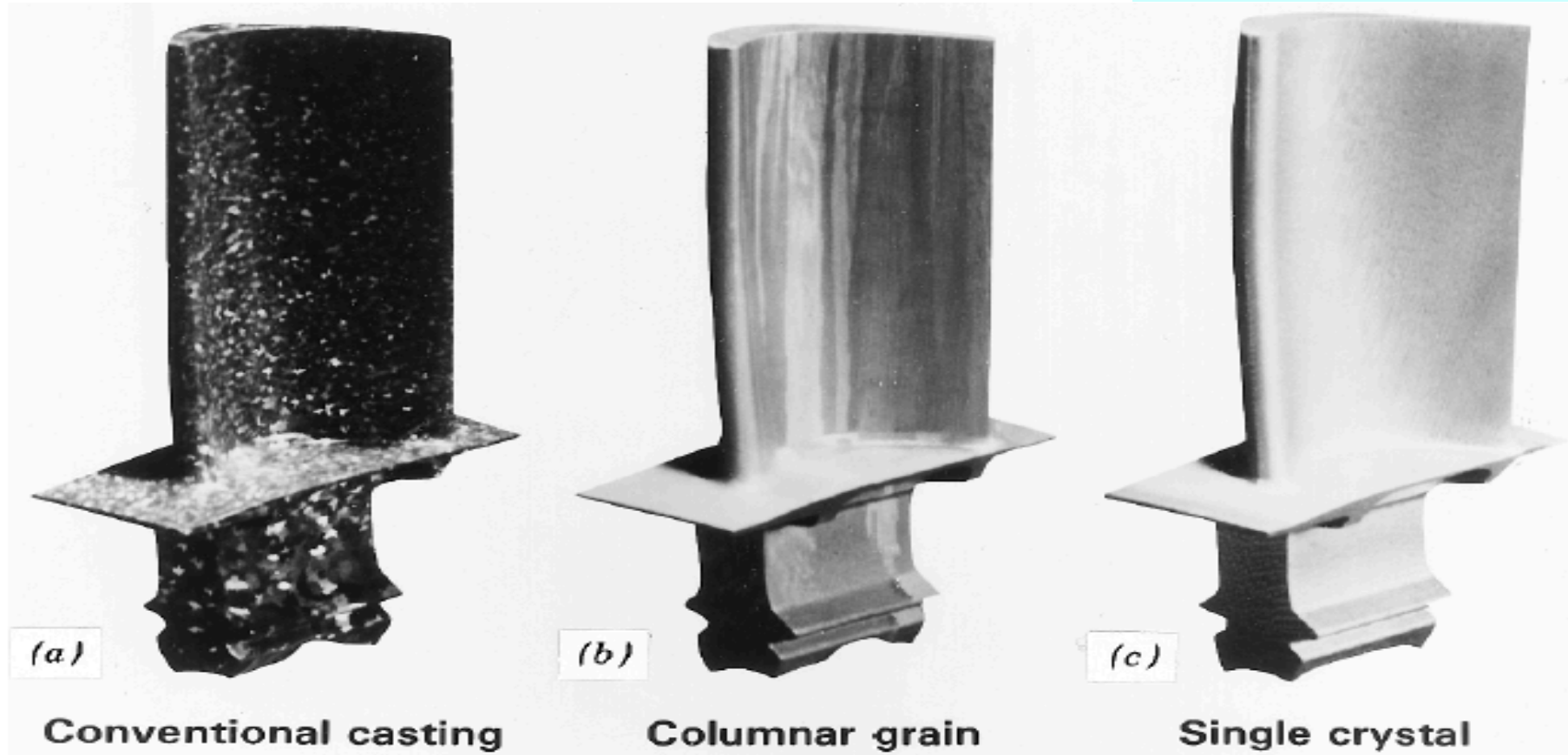
Adapted from Fig. 4.12(b), *Callister 6e*.
(Fig. 4.12(b) is courtesy of L.C. Smith and C. Brady, the National Bureau of Standards, Washington, DC [now the National Institute of Standards and Technology, Gaithersburg, MD].)

Improvement in Creep Resistance of Turbine Blades through Casting Technologies

Polycrystalline turbine blade

Columnar grain structure produced by a directional solidification technique

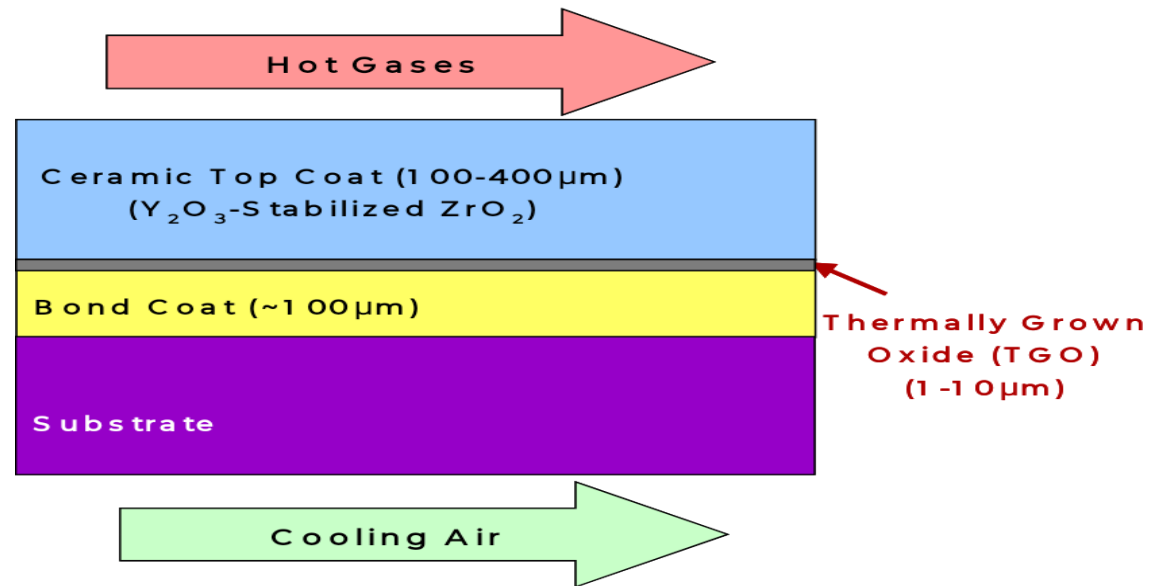
Creep resistance is further enhanced with single-crystal blades.



Thermal Barrier Coatings (TBCs)

- Demands for higher efficiency and lower emission require higher operating temperatures in aeroengines
- The typical melting points of the superalloys used for the turbine components range from 1230-1315°C
- The temp. in a combustion gas environment is $> 1370^{\circ}\text{C}$

The key to meeting the higher temperature requirements lies in providing an insulating ceramic thermal barrier coating (TBC) to lower the surface temperature of superalloy underneath



Types of Imperfections

- Vacancy atoms
- Interstitial atoms
- Substitutional atoms

Point defects

- Dislocations

Line defects

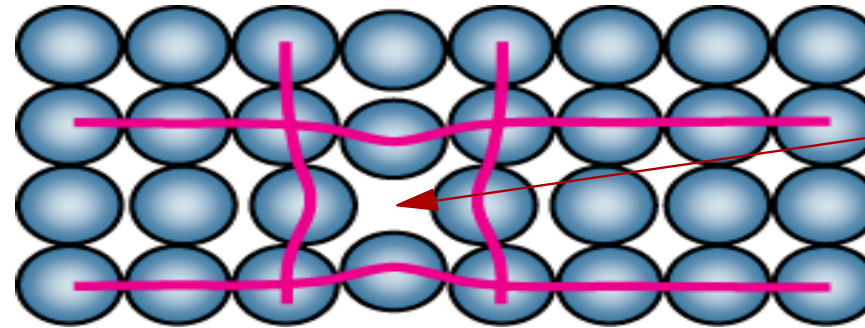
- Grain Boundaries

Area defects

Point Defects

-vacant atomic sites in a structure.

distortion
of planes

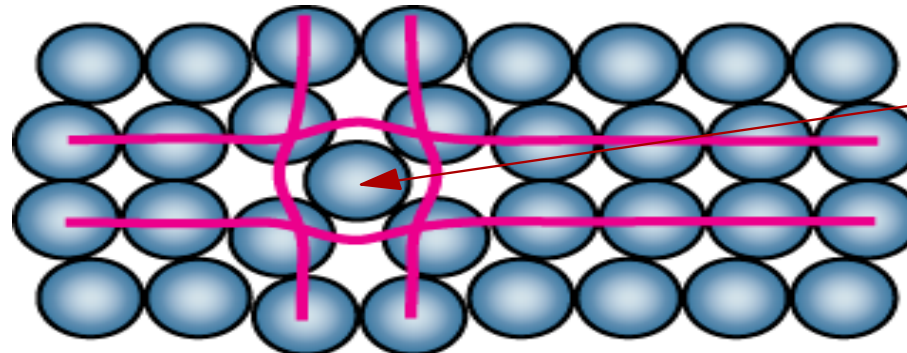


Vacancy

Self-Interstitials:

- "extra" atoms positioned between atomic sites.

distortion
of planes



self-
interstitial

Equilibrium Concentration: Point Defects

- Equilibrium concentration varies with temperature!

No. of defects $\rightarrow N_v$

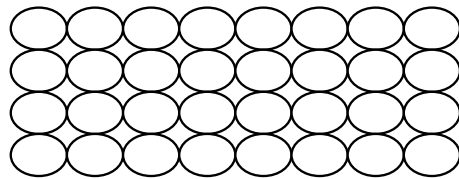
No. of potential defect sites. $\rightarrow N$

$$\frac{N_v}{N} = \exp \left(\frac{-Q_v}{kT} \right)$$

Activation energy $\rightarrow Q_v$

Boltzmann's constant $\rightarrow k$

Temperature $\rightarrow T$



Each lattice site
is a potential
vacancy site

Boltzmann's constant

$(1.38 \times 10^{-23} \text{ J/atom-K})$

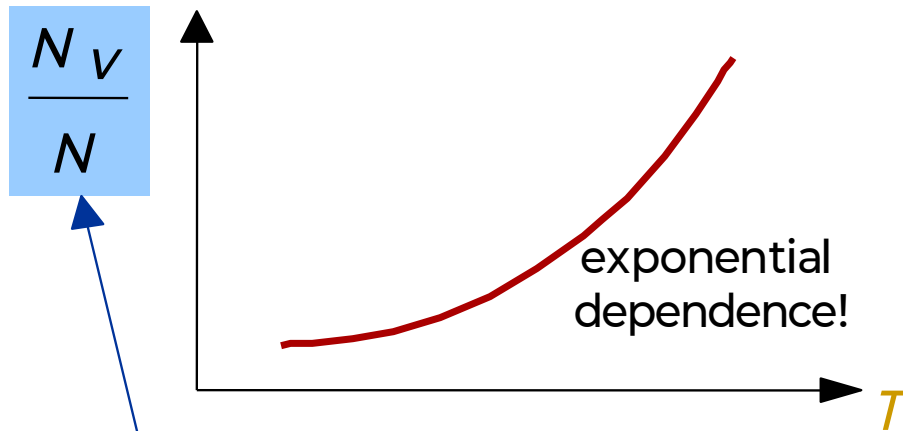
$(8.62 \times 10^{-5} \text{ eV/atom-K})$

Measuring Activation Energy

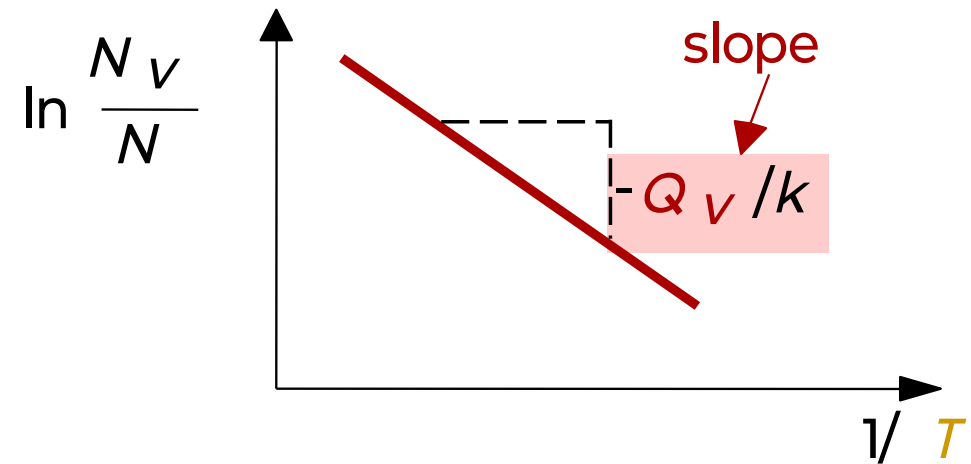
- We can get Q_v from an experiment.
- Measure this...

$$\frac{N_v}{N} = \exp \left(\frac{-Q_v}{kT} \right)$$

- Replot it...



defect concentration



Estimating Vacancy Concentration

- Find the equil. # of vacancies in 1 m³ of Cu at 1000°C.

- Given:

$$\rho = 8.4 \text{ g/cm}^3$$

$$A_{\text{Cu}} = 63.5 \text{ g/mol}$$

$$Q_v = 0.9 \text{ eV/atom}$$

$$N_A = 6.02 \times 10^{23} \text{ atoms/mol}$$

$$\frac{N_v}{N} = \exp \left(\frac{-Q_v}{kT} \right) = 2.7 \times 10^{-4}$$

Annotations for the equation above:

- Q_v is labeled 0.9 eV/atom (red arrow)
- kT is labeled 1273K (yellow arrow)
- k is labeled $8.62 \times 10^{-5} \text{ eV/atom-K}$ (green arrow)

For 1 m³, $N = \rho \times \frac{N_A}{A_{\text{Cu}}} \times 1 \text{ m}^3 = 8.0 \times 10^{28} \text{ sites}$

- Answer:

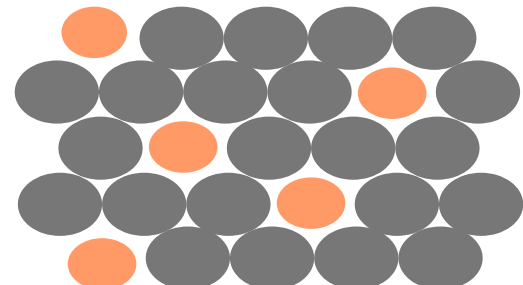
$$N_v = (2.7 \times 10^{-4})(8.0 \times 10^{28}) \text{ sites} = 2.2 \times 10^{25} \text{ vacancies}$$

06/08/2012

Point Defects in Alloys

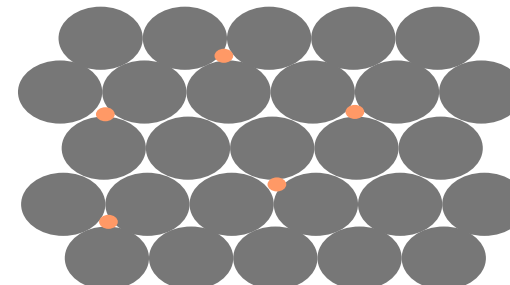
Two outcomes if impurity (B) added to host (A):

- **Solid solution** of B in A (i.e., random dist. of point defects)



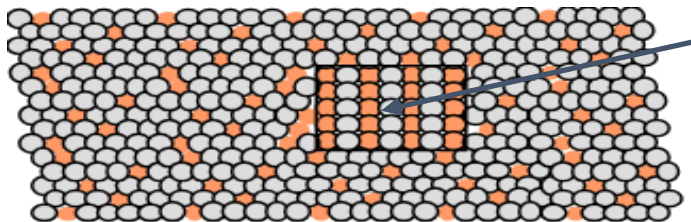
Substitutional solid soln.
(e.g., **Cu** in Ni)

OR



Interstitial solid soln.
(e.g., **C** in Fe)

- Solid solution of B in A plus particles of a new phase (usually for a larger amount of B)



Second phase particle
--different **composition**
--often different structure.

Imperfections in Solids

Conditions for substitutional solid solution (S.S.)

- W. Hume – Rothery rule
 - 1. Δr (atomic radius) < 15%
 - 2. Proximity in periodic table
 - i.e., similar electronegativities
 - 3. Same crystal structure for pure metals
 - 4. Valency
 - All else being equal, a metal will have a greater tendency to dissolve a metal of higher valency than one of lower valency

Imperfections in Solids

Application of Hume–Rothery rules – Solid Solutions

1. Would you predict more Al or Ag to dissolve in Zn?

2. More Zn or Al in Cu?

<i>Element</i>	<i>Atomic Radius (nm)</i>	<i>Crystal Structure</i>	<i>Electro-negativity</i>	<i>Valence nega-</i>
Cu	0.1278	FCC	1.9	+2
C	0.071			
H	0.046			
O	0.060			
Ag	0.1445	FCC	1.9	+1
Al	0.1431	FCC	1.5	+3
Co	0.1253	HCP	1.8	+2
Cr	0.1249	BCC	1.6	+3
Fe	0.1241	BCC	1.8	+2
Ni	0.1246	FCC	1.8	+2
Pd	0.1376	FCC	2.2	+2
Zn	0.1332	HCP	1.6	+2

Imperfections in Solids

Linear Defects (**Dislocations**)

- Are one-dimensional defects around which atoms are misaligned

Edge dislocation:

- extra half-plane of atoms inserted in a crystal structure
- $\mathbf{b} \perp$ to dislocation line

Screw dislocation:

- spiral planar ramp resulting from shear deformation
- $\mathbf{b} \parallel$ to dislocation line

Burger's vector, $\mathbf{b}$: measure of lattice distortion

Imperfections in Solids

Edge Dislocation

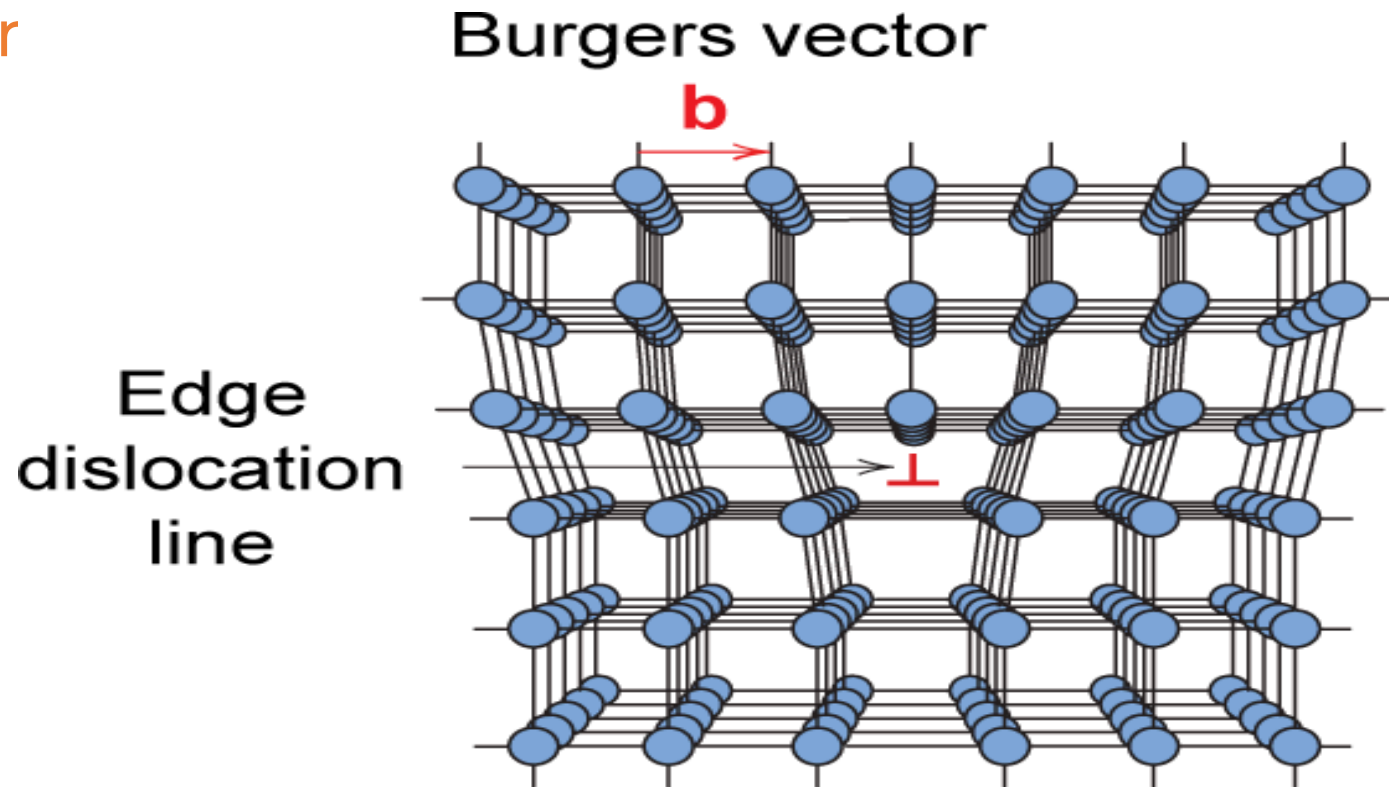
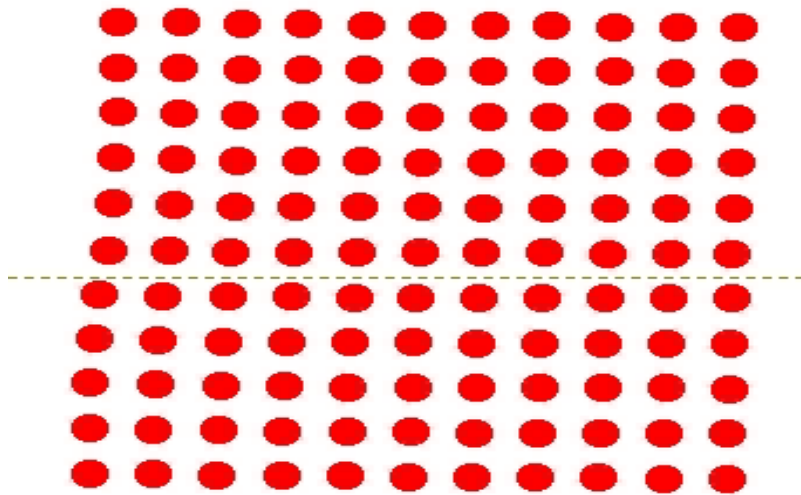


Fig. 5.7
*Callister's Materials Science and Engineering,
Adapted Version.*

Motion of Edge Dislocation

- Dislocation motion requires the successive bumping of a half plane of atoms (from left to right here).
- Bonds across the slipping planes are broken and remade in succession.



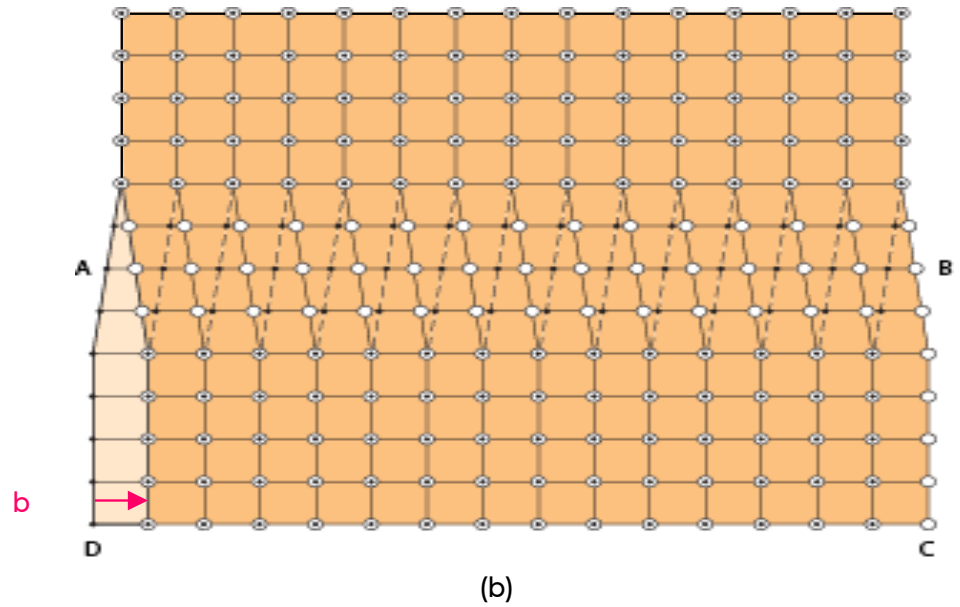
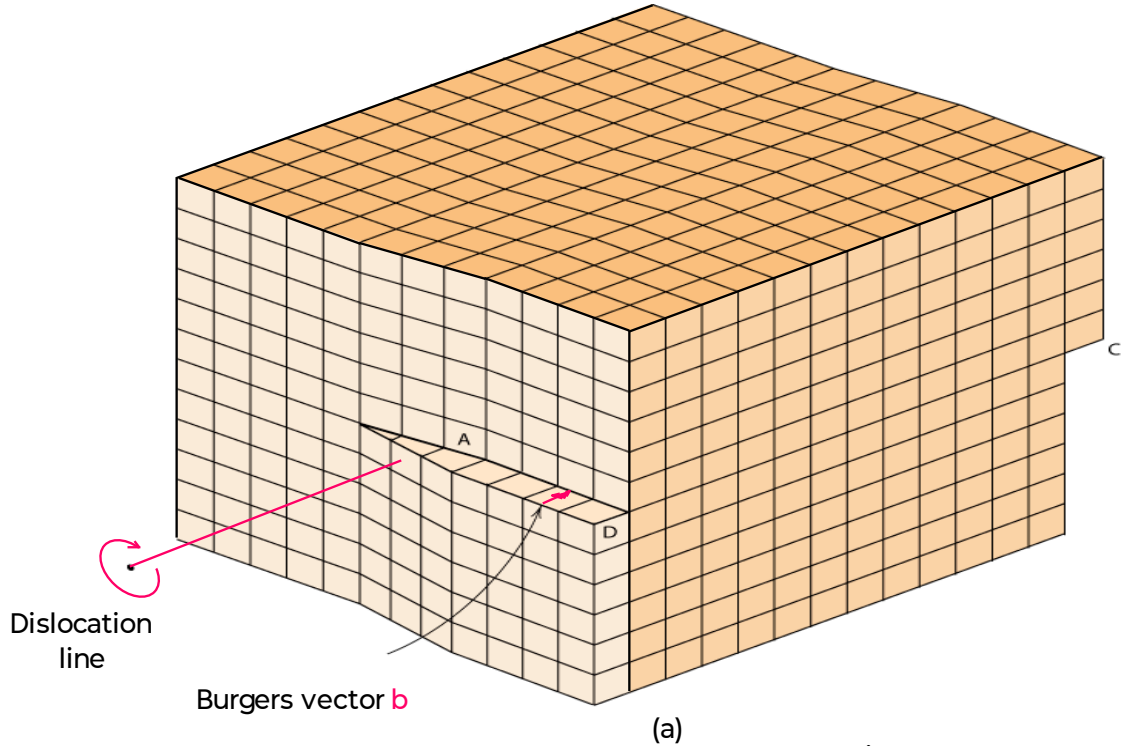
Atomic view of edge dislocation motion from left to right as a crystal is sheared.

(Courtesy P.M. Anderson)

Imperfections in Solids

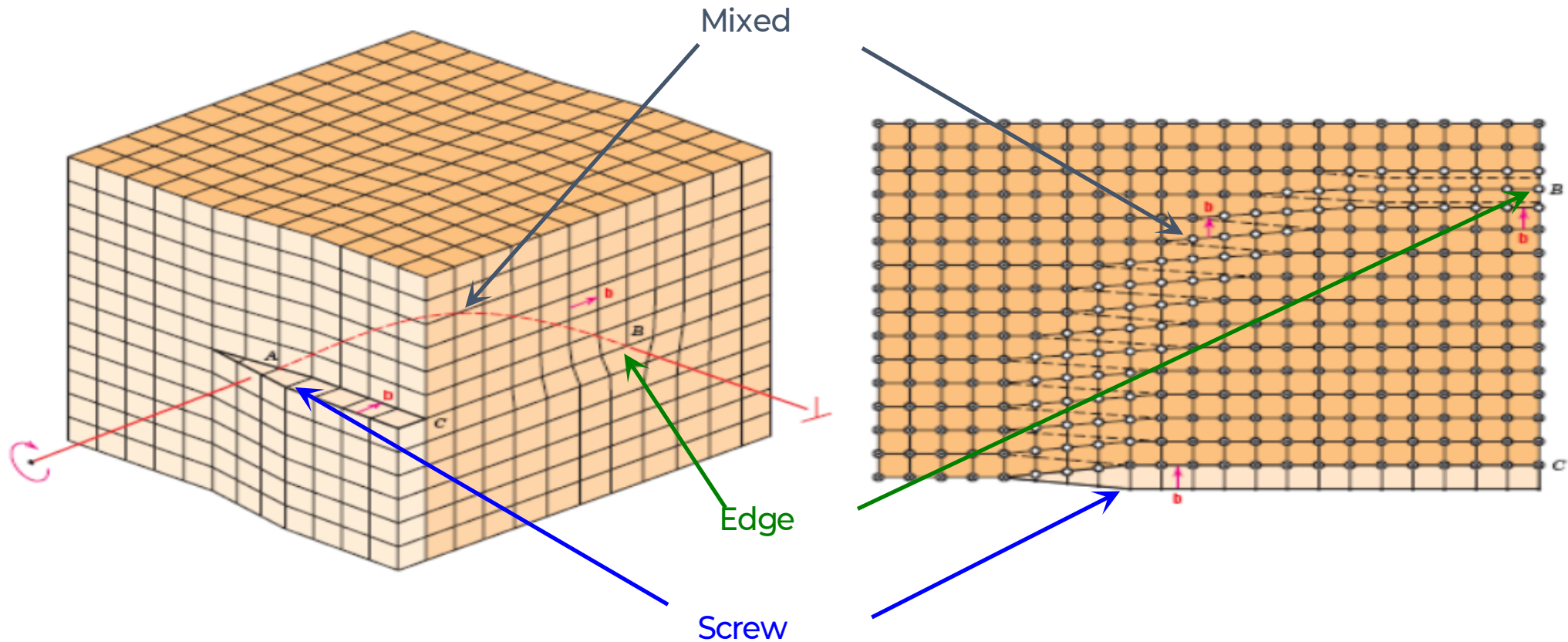


Screw Dislocation



From Fig. 5.8
Callister's Materials Science and Engineering,
Adapted Version.

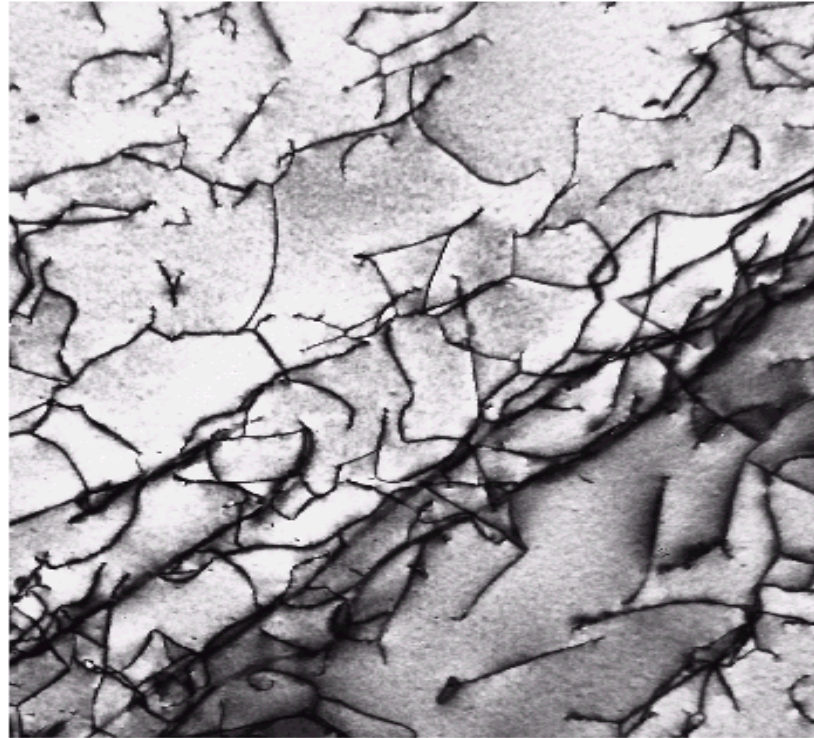
Edge, Screw, and Mixed Dislocations



From Fig. 5.9
Callister's Materials Science and Engineering,
 Adapted Version.

Imperfections in Solids

Dislocations are visible in electron micrographs



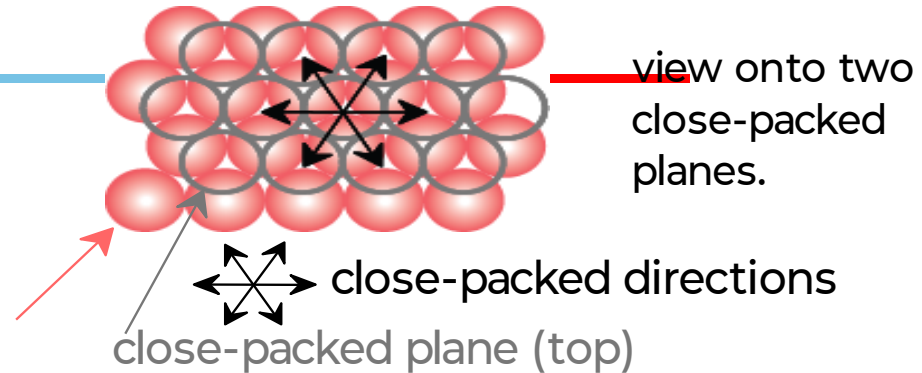
From Fig. 5.10
Callister's Materials Science and Engineering,
Adapted Version.

Dislocations & Crystal Structures



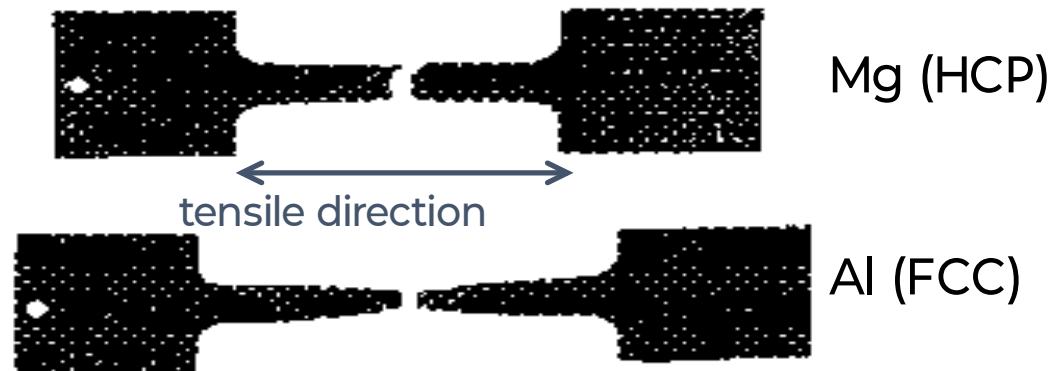
- Structure: **close-packed** planes & directions are preferred.

close-packed plane (bottom)



- Comparison among crystal structures:
FCC: many close-packed planes/directions;
HCP: only one plane, 3 directions;
BCC: none

- Specimens that were tensile tested.



Summary

- Point, Line, and Area defects exist in solids.
- The number and type of defects can be varied and controlled (e.g., T controls vacancy conc.)
- Defects affect material properties (e.g., grain boundaries control crystal slip).
- Defects may be desirable or undesirable (e.g., dislocations may be good or bad, depending on whether plastic deformation is desirable or not.)

Some Important Points:



Determine the number of lattice points per cell in the cubic crystal systems. If there is only one atom located at each lattice point, calculate the number of atoms per unit cell.

In the SC unit cell, lattice points are located only at the corners of the cube:

$$\frac{\text{lattice point}}{\text{unit cell}} = (8 \text{ corners}) \left(\frac{1}{8} \right) = 1$$

In BCC unit cells, lattice points are located at each corner and with one at the center of the cube:

$$\frac{\text{lattice point}}{\text{unit cell}} = (8 \text{ corners}) \left(\frac{1}{8} \right) + (1 \text{ center})(1) = 2$$

In FCC unit cells, lattice points are located at all corners and all faces of the cube:

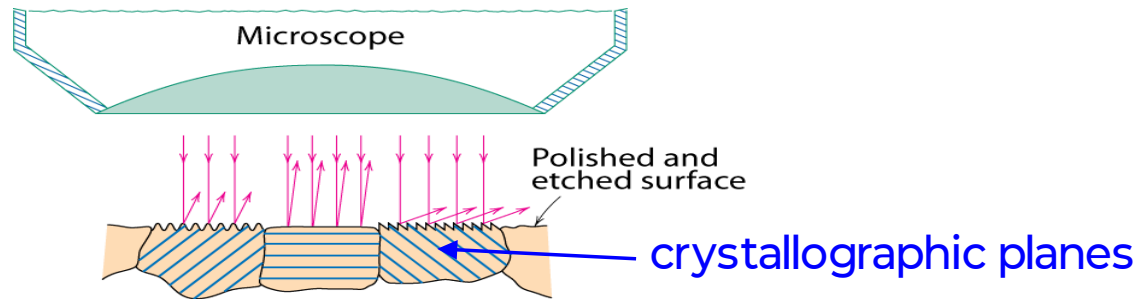
$$\frac{\text{lattice point}}{\text{unit cell}} = (8 \text{ corners}) \left(\frac{1}{8} \right) + (6 \text{ faces}) \left(\frac{1}{2} \right) = 4$$

Microscopic Examination

- Crystallites (grains) and grain boundaries. Vary considerably in size. Can be quite large
 - ex: Large single crystal of quartz or diamond or Si
 - ex: Aluminum light post or garbage can - see the individual grains
- Crystallites (grains) can be quite small (mm or less) – necessary to observe with a microscope.

Optical Microscopy

- Useful up to 2000X magnification.
- Polishing removes surface features (e.g., scratches)
- Etching changes reflectance, depending on crystal orientation.



From Fig. 5.18(b) and (c)
Callister's Materials Science and Engineering,
Adapted Version.

(Fig. 5.18(c) is courtesy
 of J.E. Burke, General Electric Co.



Micrograph of
 brass (a Cu-Zn alloy)

Optical Microscopy

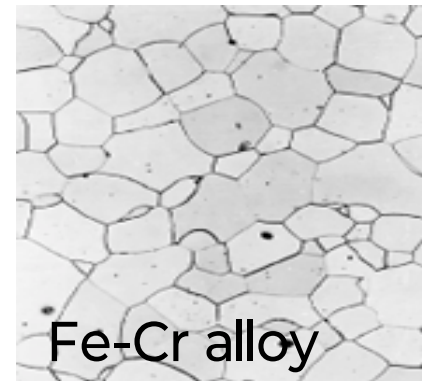
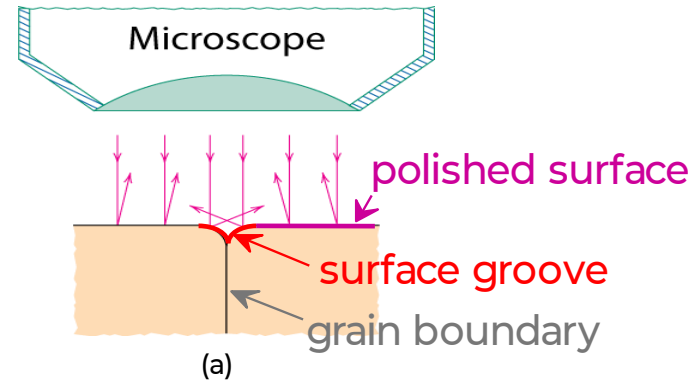
Grain boundaries...

- are imperfections,
- are more susceptible to etching,
- may be revealed as dark lines,
- change in crystal orientation across boundary.

ASTM grain size number

$$N = 2^{n-1}$$

number of grains/in²
at 100x
magnification



(b)

From Fig. 5.19(a) and (b)
Callister's Materials Science and Engineering, Adapted Version.

(Fig. 5.19(b) is courtesy of L.C. Smith and C. Brady, the National Bureau of Standards, Washington, DC [now the National Institute of Standards and Technology, Gaithersburg, MD].)

Optical Microscopy

- Polarized light
 - metallographic scopes often use polarized light to increase contrast
 - Also used for transparent samples such as polymers

Microscopy

Optical resolution ca. 10^{-7} m = $0.1\text{ }\mu\text{m}$ = 100 nm

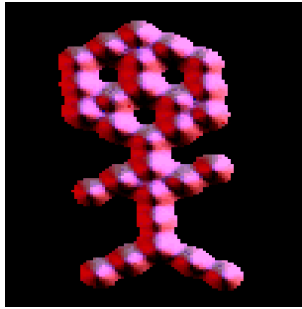
For higher resolution need higher frequency

- X-Rays? Difficult to focus.
- Electrons
 - wavelengths ca. 3 pm (0.003 nm)
 - (Magnification - 1,000,000X)
 - Atomic resolution possible
 - Electron beam focused by magnetic lenses.

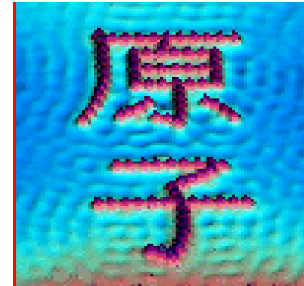
Scanning Tunneling Microscopy (STM)



- Atoms can be arranged and imaged!



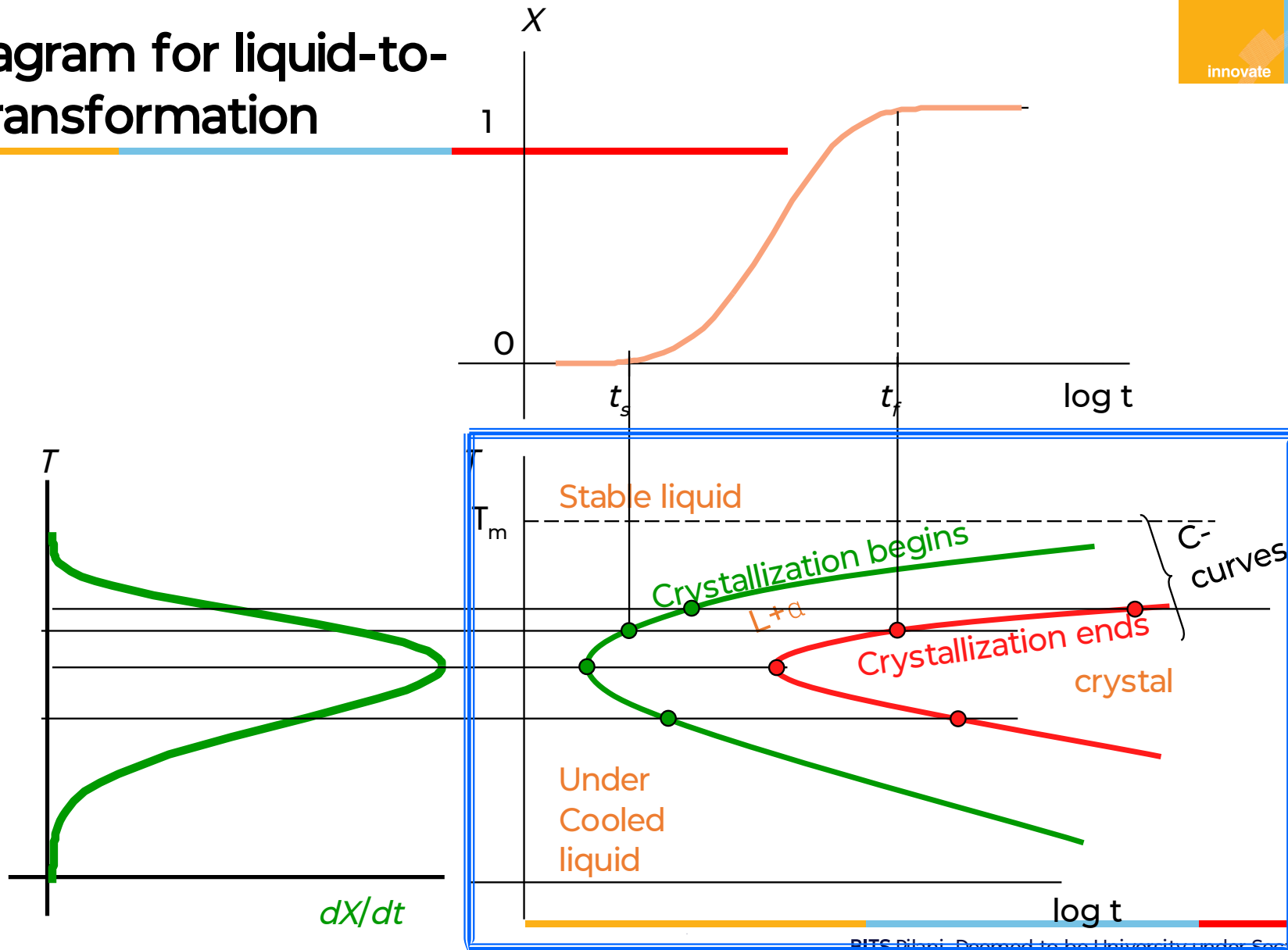
Carbon monoxide molecules arranged on a platinum (111) surface.



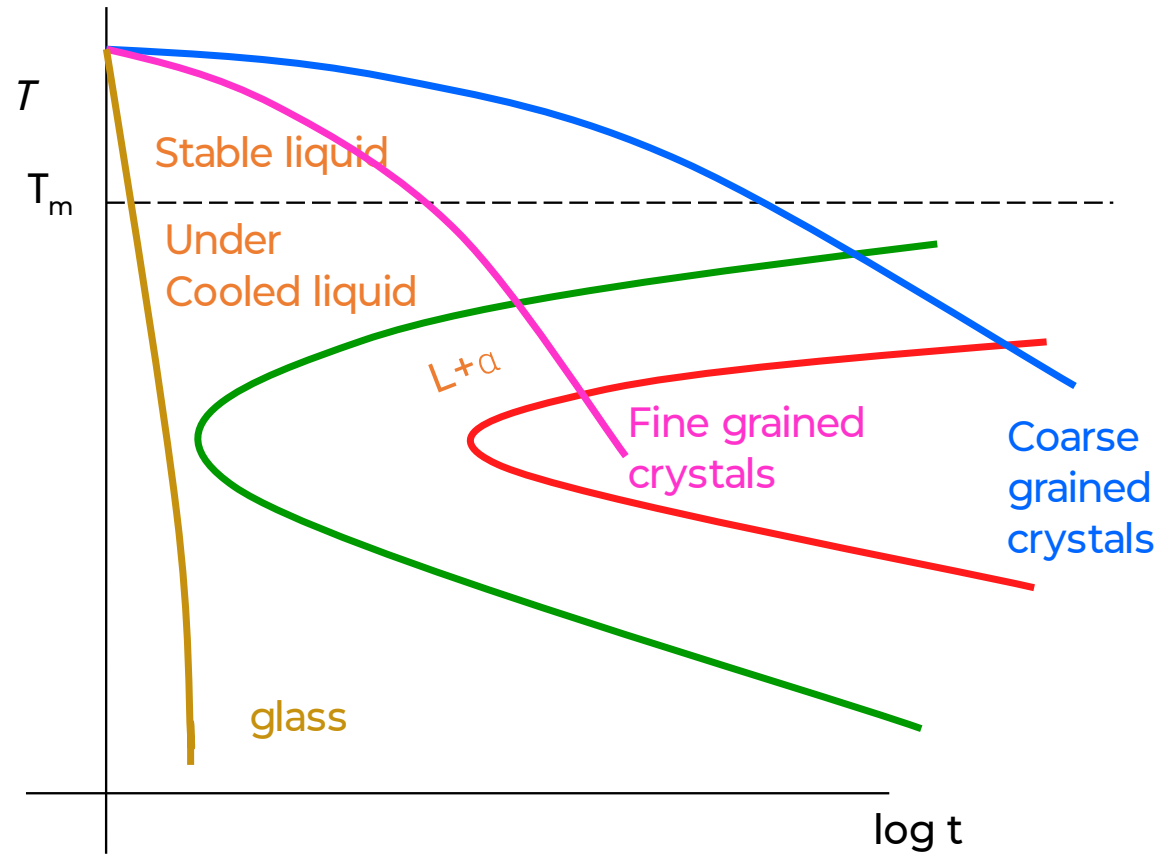
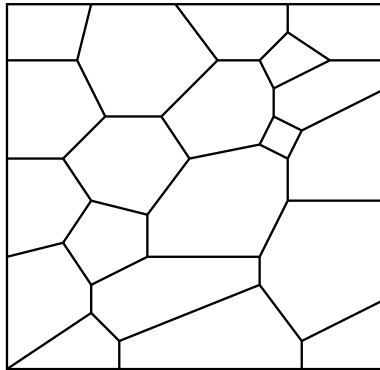
Iron atoms arranged on a copper (111) surface. These Kanji characters represent the word “atom”.

Photos produced from the work of C.P. Lutz, Zeppenfeld, and D.M. Eigler. Reprinted with permission from International Business Machines Corporation, copyright 1995.

TTT Diagram for liquid-to-solid transformation

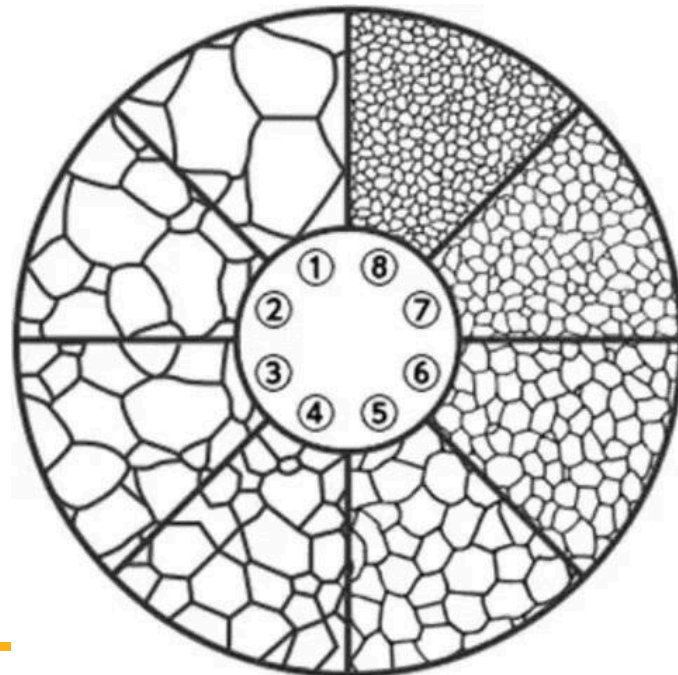


TTT Diagram for liquid-to-solid transformation



Grain structure....**ASTM Standard E112**

Grain-Size Number	Mean Number of Grains per Square Inch (at 100x Magnification)
1	1
3	4
6	32
10	512
12	2,048
14	8,192



Thank You



BITS Pilani

Pilani | Dubai | Goa | Hyderabad

DE ZG 631 Materials Technology & Testing

Atomic Structure



Atom – Electrons – 9.11×10^{-31} kg
 Protons } 1.67×10^{-27} kg
 Neutrons }

Atomic Number = No. of **protons** in nucleus of atom
= No. of **electrons** of *neutral species*

A [=] **atomic mass unit** = amu = $1/12$ mass of ^{12}C (C = 12.011)

Atomic wt = wt of 6.023×10^{23} molecules or atoms
1 amu/atom = 1g/mol

Electronic Structure



Electrons have wavelike and particulate properties.

- This means that electrons are in **orbitals** defined by a probability.
- Each orbital at discrete energy level determined by **quantum numbers**.

Quantum

n = principal (energy level-shell)
etc.)

l = subsidiary (orbitals)

m = magnetic

Designation

K, L, M, N, O (1, 2, 3,

s, p, d, f (0, 1, 2, 3, ..., $n-1$)

SURVEY OF ELEMENTS



Element	Atomic #	Electron configuration
Hydrogen	1	$1s^1$
Helium	2	$1s^2$ (stable)
Lithium	3	$1s^2 2s^1$
Beryllium	4	$1s^2 2s^2$
Boron	5	$1s^2 2s^2 2p^1$
Carbon	6	$1s^2 2s^2 2p^2$
...	...	...
Neon	10	$1s^2 2s^2 2p^6$ (stable)
Sodium	11	$1s^2 2s^2 2p^6 3s^1$
Magnesium	12	$1s^2 2s^2 2p^6 3s^2$
Aluminum	13	$1s^2 2s^2 2p^6 3s^2 3p^1$
...	...	...
Argon	18	$1s^2 2s^2 2p^6 3s^2 3p^6$ (stable)
...	...	...
Krypton	36	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$ (stable)

- Most elements: Electron configuration not stable.
- Why? Valence (outer) shell usually not filled completely.

Atomic Structure



Valence electrons determine all of the following properties

- 1) Chemical (Reactivity)
- 2) Electrical (Electron conductivity)
- 3) Thermal (Conductivity)
- 4) Optical (Transition of electrons from one electronic state to another)

Electron Configurations



Valence electrons – those in unfilled shells

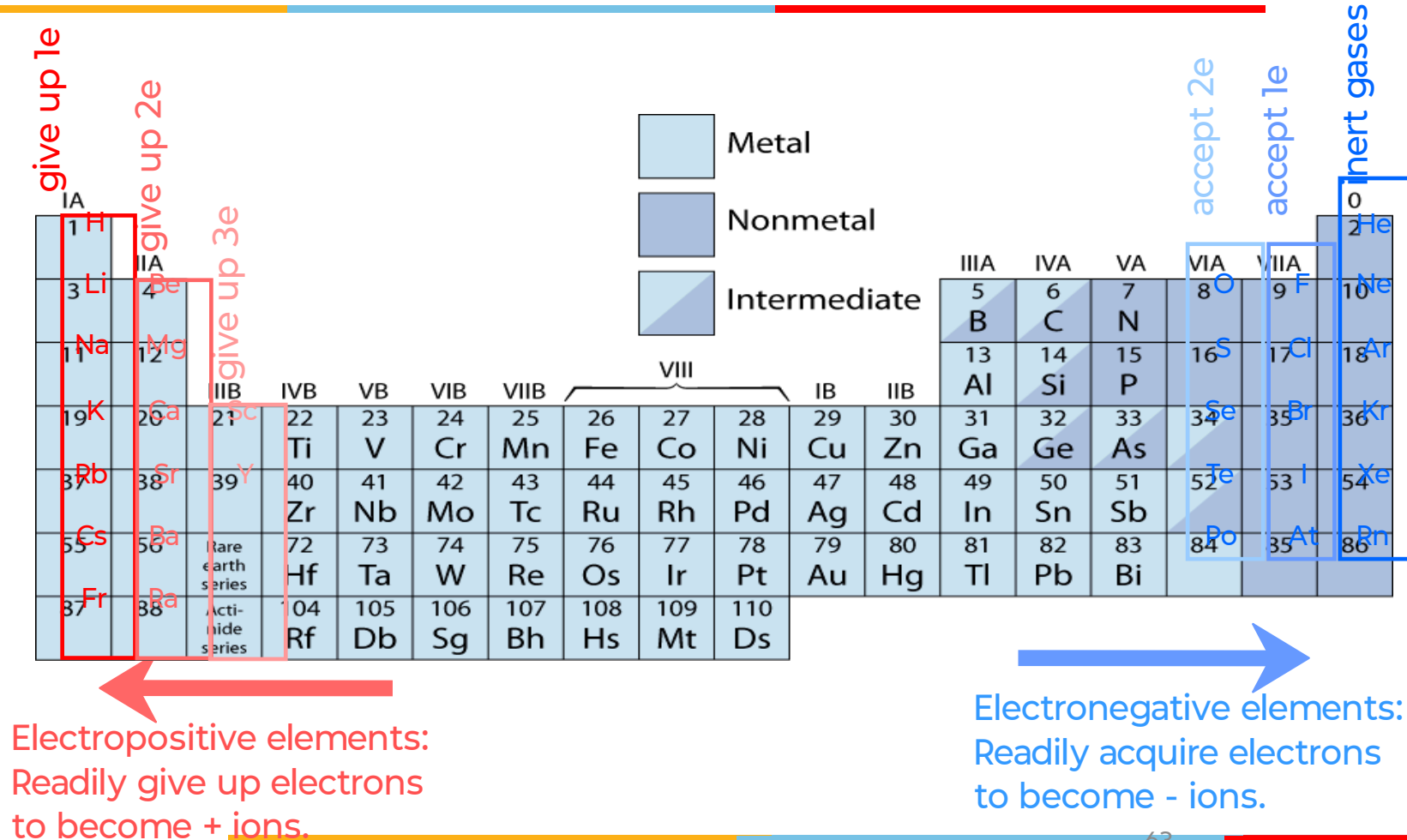
Filled shells more stable

Valence electrons are most available for bonding and tend to control the chemical properties

– example: C (atomic number = 6)



The Periodic Table



From Fig. 2.
Callister's
Materials
Science and
Engineering,
Adapted
Version.

Interatomic Bonding



PRIMARY INTERATOMIC BONDS:

Ionic Bonding

Covalent Bonding

Metallic Bonding

SECONDARY BONDING OR VAN DER WAALS BONDING

Fluctuating Induced Dipole Bonds

Polar Molecule-Induced Dipole Bonds

Permanent Dipole Bonds

NOTE: Hydrogen bonding, a special type of secondary bonding, is found to exist between some molecules that have hydrogen as one of the constituents.

Ionic bond – metal + nonmetal

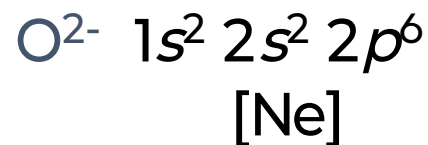
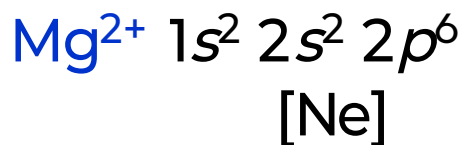
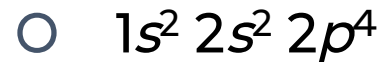
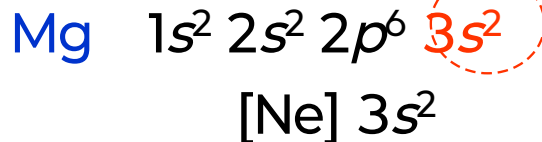


↑
donates
electrons

↑
accepts
electrons

Dissimilar electronegativities

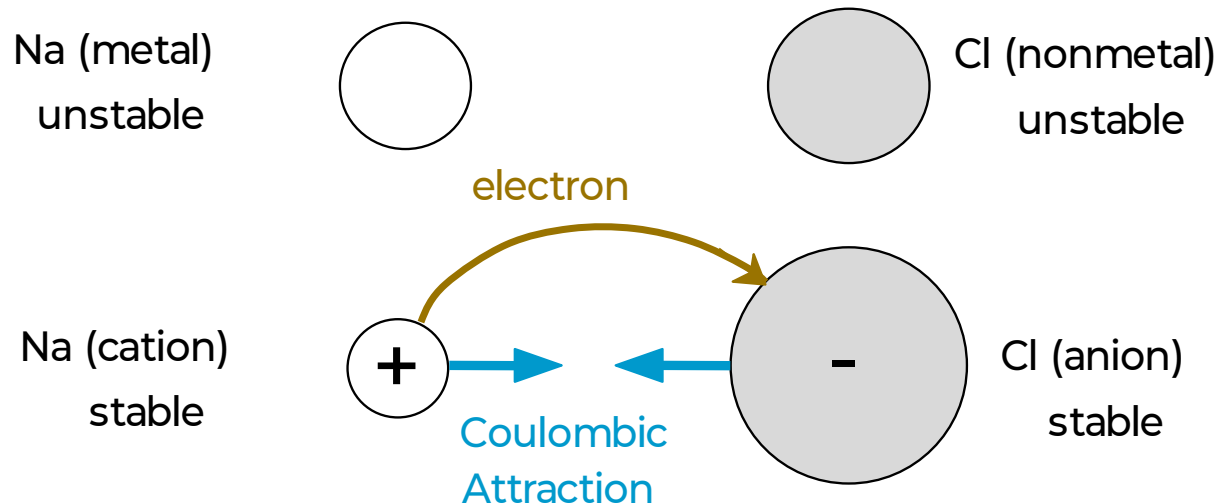
ex: MgO



Ionic Bonding



- Occurs between + and - ions.
- Requires **electron transfer**.
- Large difference in electronegativity required.
- Example: NaCl



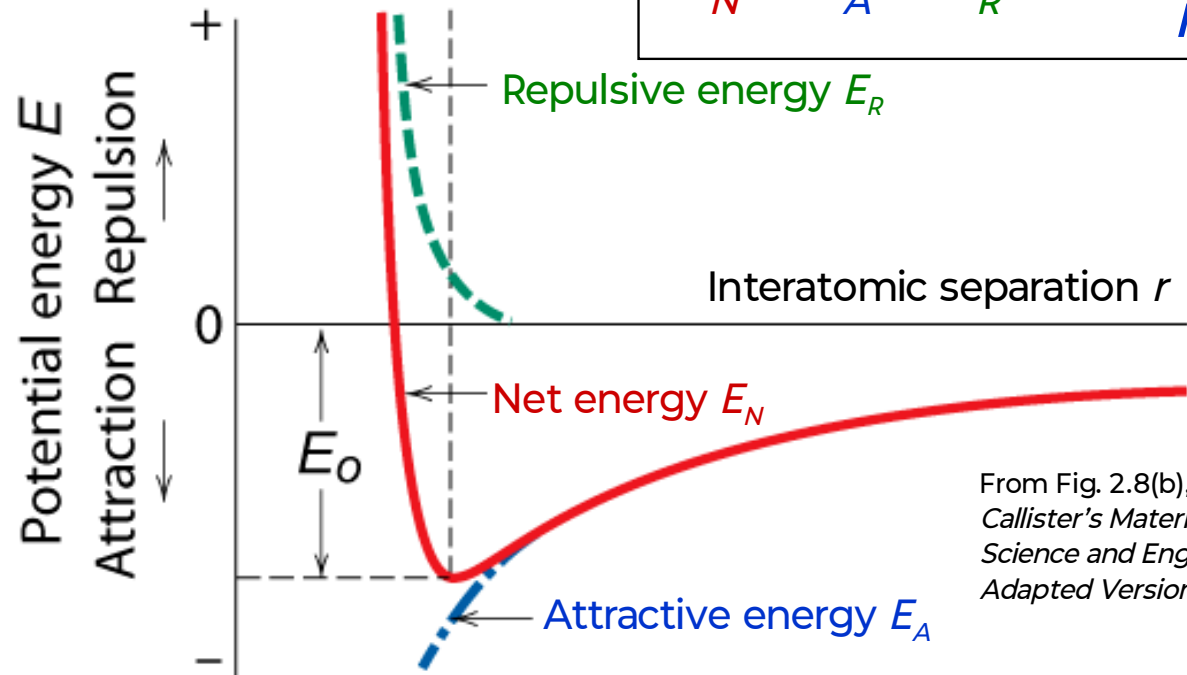
Ionic Bonding



Energy – minimum energy most stable

- Energy balance of attractive and repulsive terms

$$E_N = E_A + E_R = -\frac{A}{r} + \frac{B}{r^n}$$



From Fig. 2.8(b),
Callister's Materials
Science and Engineering,
Adapted Version.

Examples: Ionic Bonding

- Predominant bonding in **Ceramics**

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Covalent Bonding

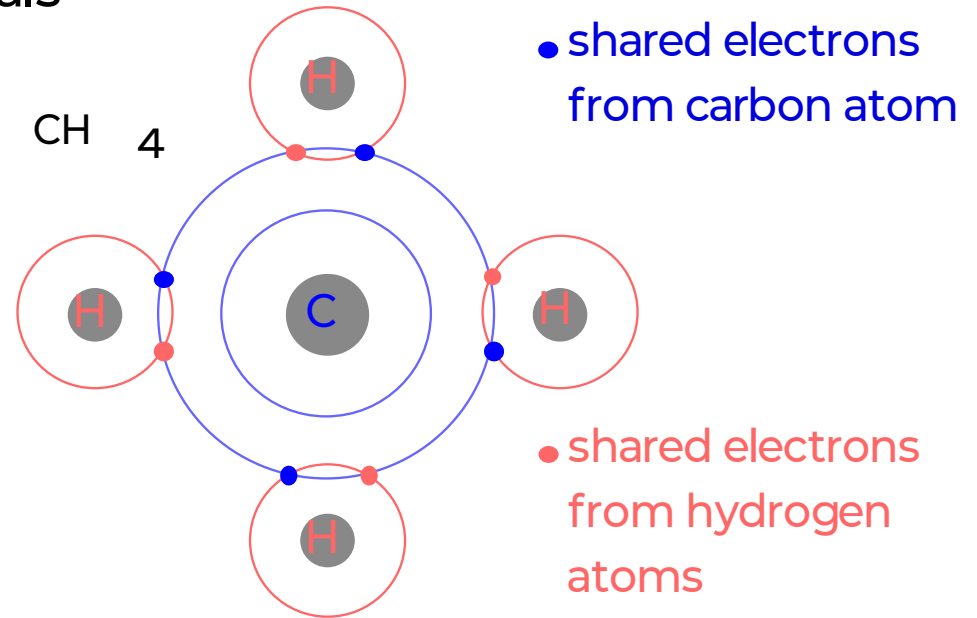


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Callister's Materials Science and Engineering, Adapted Version.

Primary Bonding



Metallic Bond -- delocalized as electron cloud

Ionic-Covalent Mixed Bonding

$$\% \text{ ionic character} = \left(1 - e^{-\frac{(X_A - X_B)^2}{4}} \right) \times (100 \%)$$

where X_A & X_B are Pauling electronegativities

Ex: MgO

$$\begin{aligned} X_{\text{Mg}} &= 1.3 \\ X_{\text{O}} &= 3.5 \end{aligned}$$

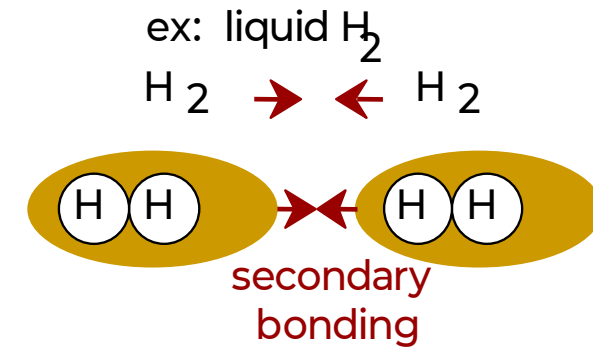
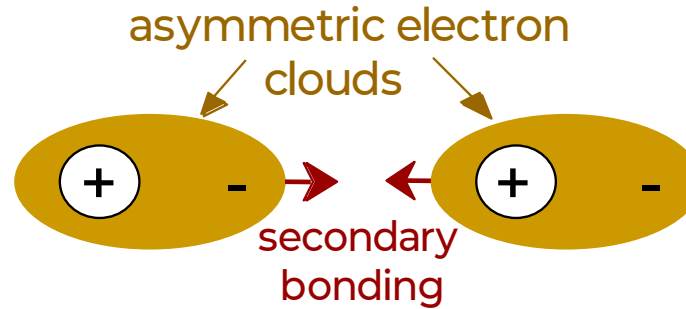
$$\% \text{ ionic character} = \left(1 - e^{-\frac{(3.5 - 1.3)^2}{4}} \right) \times (100\%) = 70.2\% \text{ ionic}$$

SECONDARY BONDING



Arises from interaction between dipoles

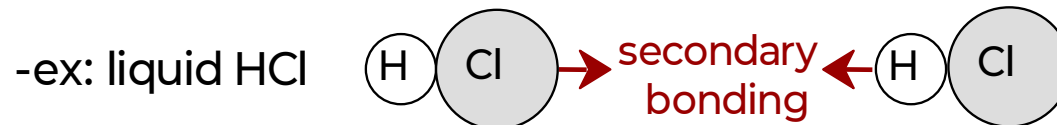
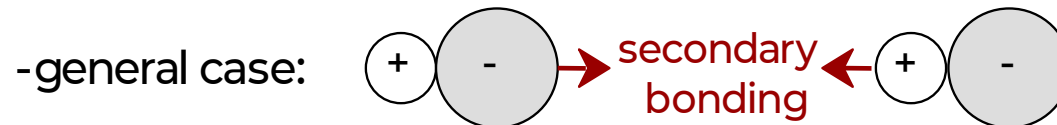
- Fluctuating dipoles



- Permanent dipoles-molecule induced

From Fig. 2.13

Callister's Materials Science and Engineering, Adapted Version.



From Fig. 2.14

*Callister's Materials Science and Engineering
Adapted Version.*



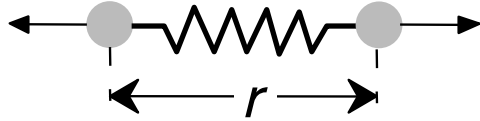
Summary: Bonding

Type	Bond Energy	Comments
Ionic	Large!	Nondirectional (ceramics)
Covalent	Variable large-Diamond small-Bismuth	Directional (semiconductors , ceramics , polymer chains)
Metallic	Variable large-Tungsten small-Mercury	Non directional (metals)
Secondary	Smallest	Directional inter-chain (polymer) inter-molecular

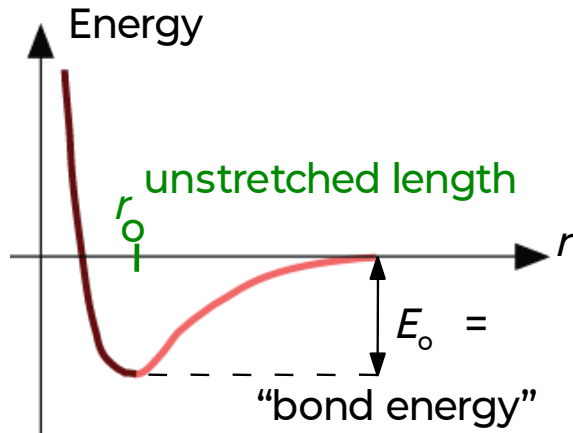
Properties From Bonding: T_m



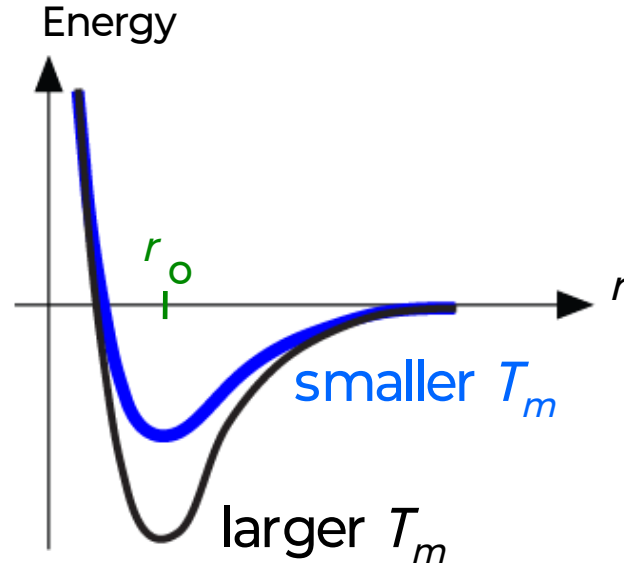
- Bond length, r



- Bond energy, E_o

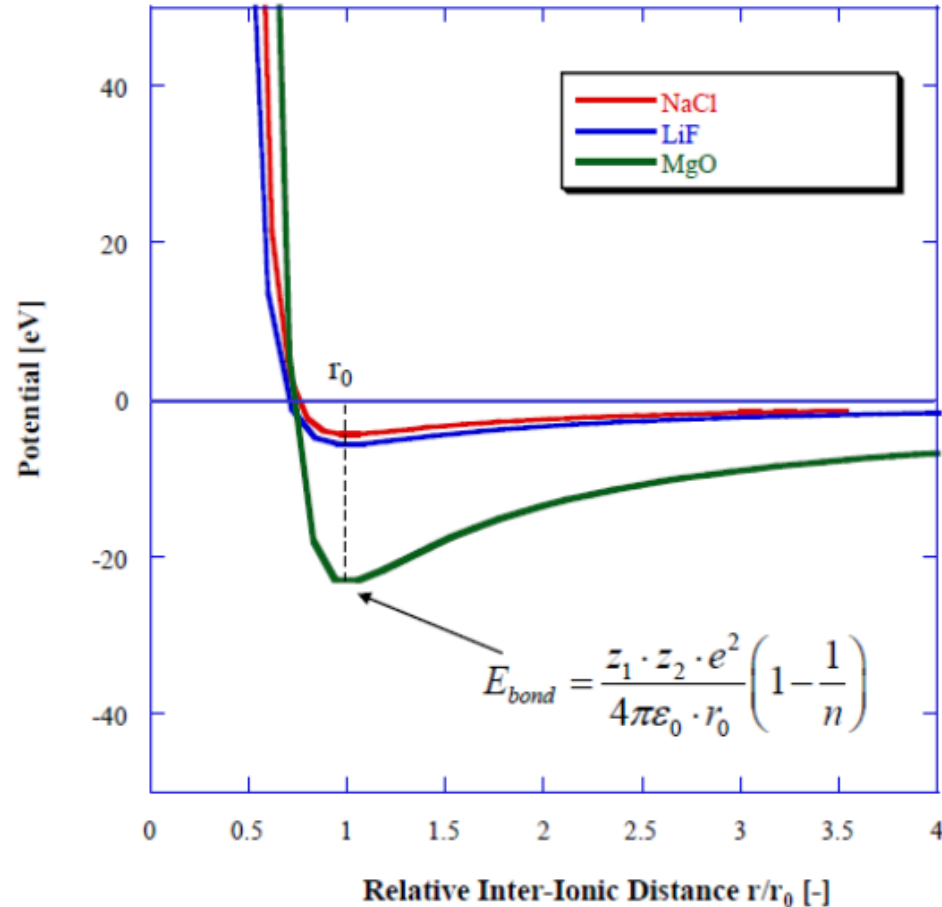


- Melting Temperature, T_m



T_m is larger if E_o is larger.

Melting Temperature: An Illustration

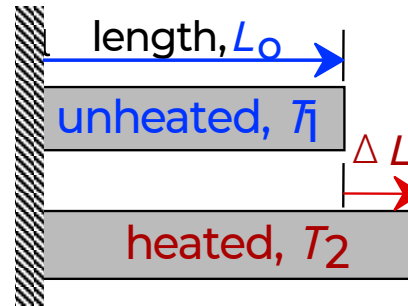


MgO	NaCl	LiF	Crystal Structure
2852°C	801°C	848°C	Rocksalt

Properties From Bonding : α



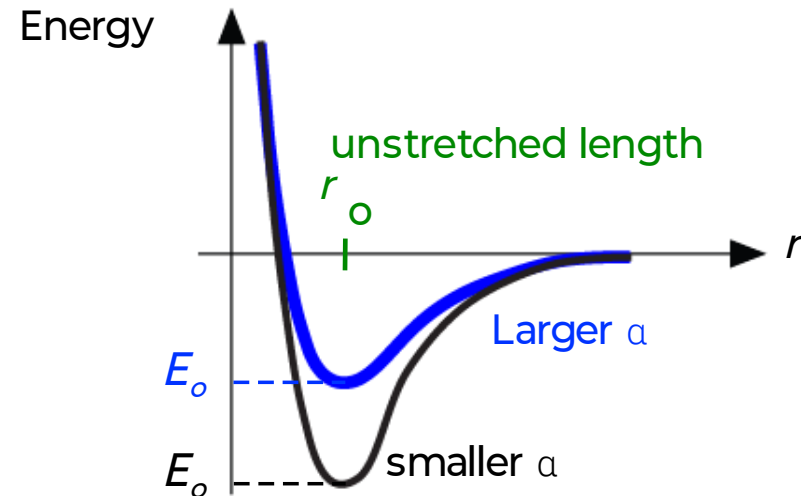
- Coefficient of thermal expansion,



coeff. thermal expansion

$$\frac{\Delta L}{L_0} = \alpha (T_2 - T_1)$$

- $\alpha \sim$ symmetry at r_0



α is larger if E_0 is smaller.

Summary:



Ceramics

(Ionic & covalent bonding):

Large bond energy

large T_m

large E

small α

Metals

(Metallic bonding):

Variable bond energy

moderate T_m

moderate E

moderate α

Polymers

(Covalent & Secondary):

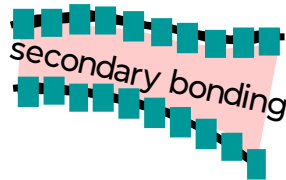
Directional Properties

Secondary bonding dominates

small T_m

small E

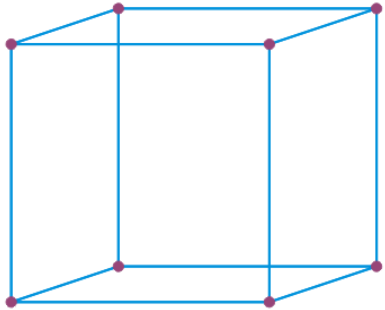
large α



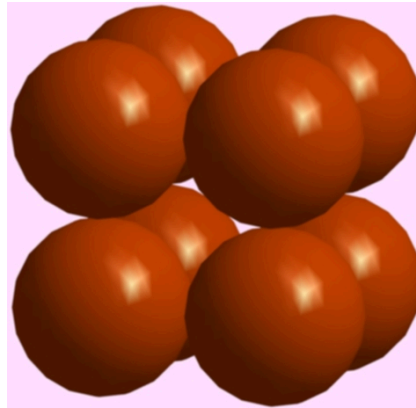
Cubic Crystal System: Simple Cubic



Simple Cubic (SC) Lattice + Sphere Motif

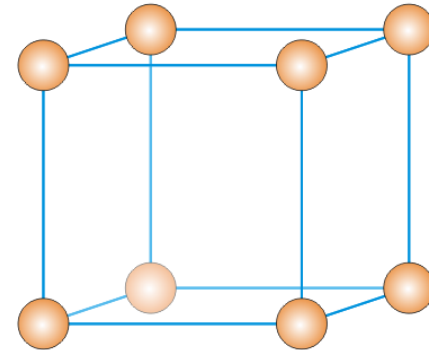


Unit cell of the SC lattice



=

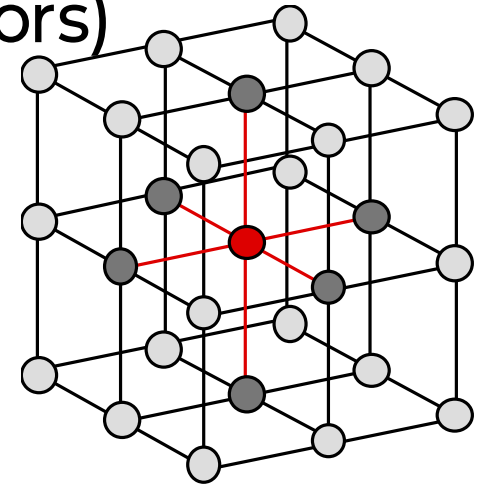
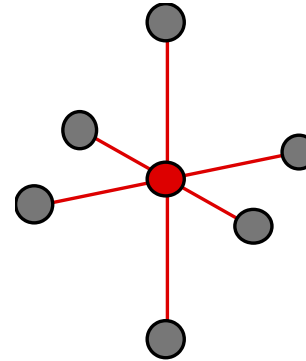
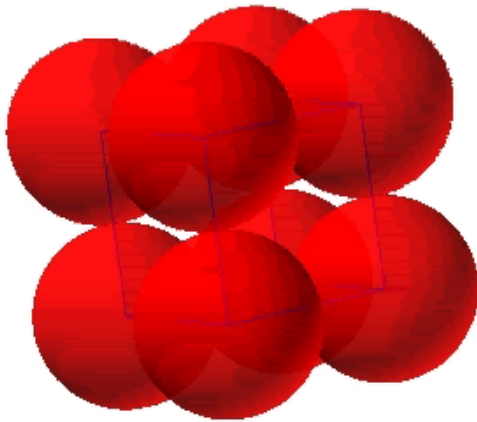
Simple Cubic Crystal



Simple Cubic Structure (SC)



- Rare due to low packing density (only Po has this structure)
- Close-packed directions are cube edges.
 - Coordination # = 6
(# nearest neighbors)



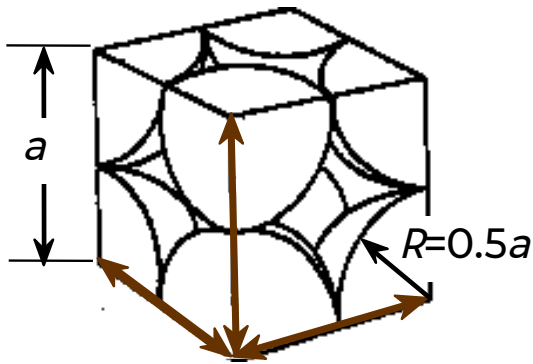
(Courtesy P.M. Anderson)

Atomic Packing Factor (APF)



$$APF = \frac{\text{Volume of atoms in unit cell}^*}{\text{Volume of unit cell}}$$

*assume hard spheres



close-packed directions

contains $8 \times 1/8 =$

1 atom/unit cell

$$APF = \frac{\text{atoms unit cell} \times \frac{4}{3} \pi (0.5a)^3}{a^3}$$

Labels in diagram:
- 'atoms unit cell' points to the '1' in the numerator.
- 'volume atom' points to the $\frac{4}{3} \pi (0.5a)^3$ term.
- 'volume unit cell' points to the a^3 term in the denominator.

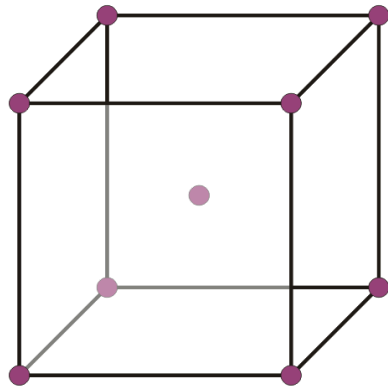
• APF for a simple cubic structure = 0

From Fig. 4.38

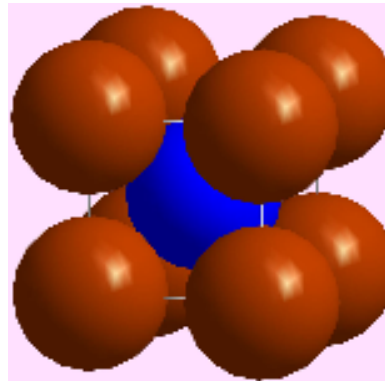
Callister's Materials Science and Engineering,
Adapted Version.

Cubic Crystal System: BCC Lattice

Body Centred Cubic (BCC) Lattice+ Sphere Motif

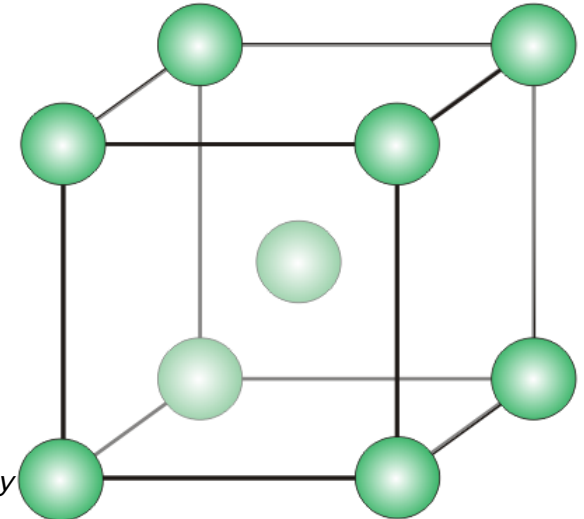


Unit cell of the BCC lattice



Central atom is coloured differently for better visibility

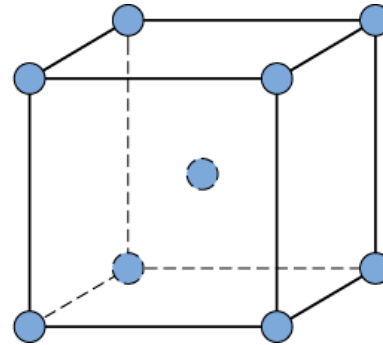
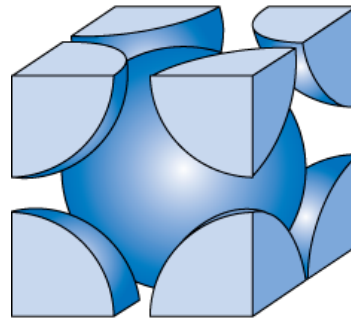
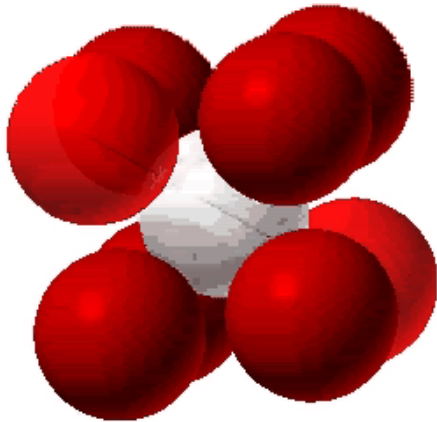
Body Centred Cubic Crystal



Body Centered Cubic Structure (BCC)



- Atoms touch each other along cube diagonals.
 - Note: All atoms are identical; the center atom is shaded differently only for ease of viewing.
- ex: Cr, W, Fe (α), Tantalum, Molybdenum
- Coordination # = 8

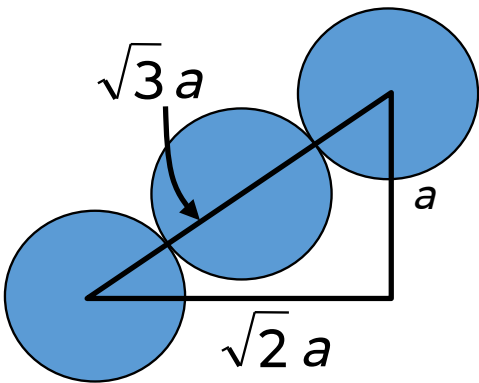
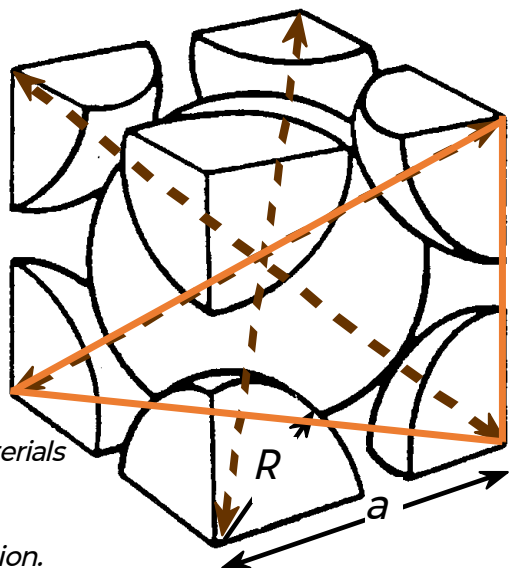


From Fig. 4.2,
Callister's Materials Science and Engineering,
Adapted Version.

(Courtesy P.M. Anderson)

2 atoms/unit cell: 1 center + 8 corners $\times$ 1/8

Atomic Packing Factor: BCC



Close-packed directions:
length = $4R = \sqrt{3} a$

From
Fig. 4.2(a)
Callister's Materials
Science and
Engineering,
Adapted Version.

atoms

unit cell

APF =

2

$\frac{4}{3} \pi \left(\frac{\sqrt{3} a}{4} \right)^3$

volume

atom

a^3

volume

unit cell

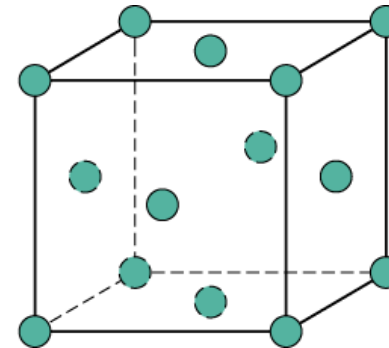
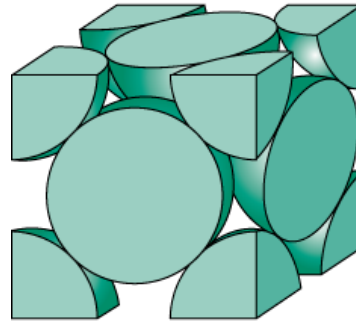
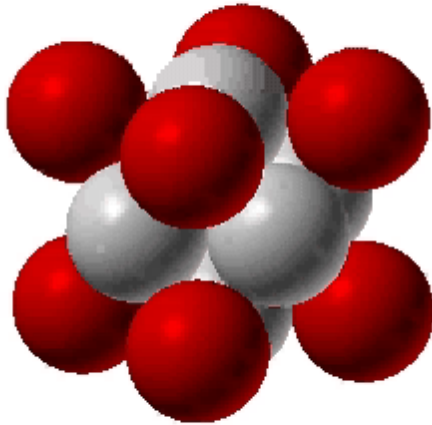
• APF for a body-centered cubic structure = 0.68

Cubic Crystal System: Face Centered Cubic Structure (FCC)



- Atoms touch each other along face diagonals.
 - Note: All atoms are identical; the face-centered atoms are shaded differently only for ease of viewing.
ex: Al, Cu, Au, Pb, Ni, Pt, Ag

- Coordination # = 12

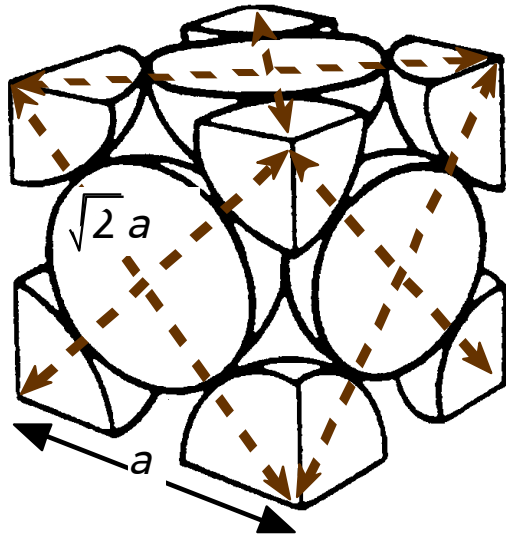


From Fig. 4.1
Callister's Materials Science and Engineering,
Adapted Version.

4 atoms/unit cell: $6 \text{ face} \times \frac{1}{2} + 8 \text{ corners} \times \frac{1}{8}$

(Courtesy P.M. Anderson)

Atomic Packing Factor: FCC



maximum achievable APF

Close-packed directions:

$$\text{length} = 4R = \sqrt{2} a$$

Unit cell contains:

$$6 \times 1/2 + 8 \times 1/8$$

$$= 4 \text{ atoms/unit cell}$$

From
Fig. 4.1(a),
Callister's Materials Science
and Engineering,
Adapted Version.

atoms
unit cell

APF =

$$\frac{4 \times \frac{4}{3} \pi \left(\frac{\sqrt{2} a}{4} \right)^3}{a^3}$$

volume
atom

volume
unit cell

• APF for a face-centered cubic structure = 0.74



BITS Pilani

Microstructure and Applications

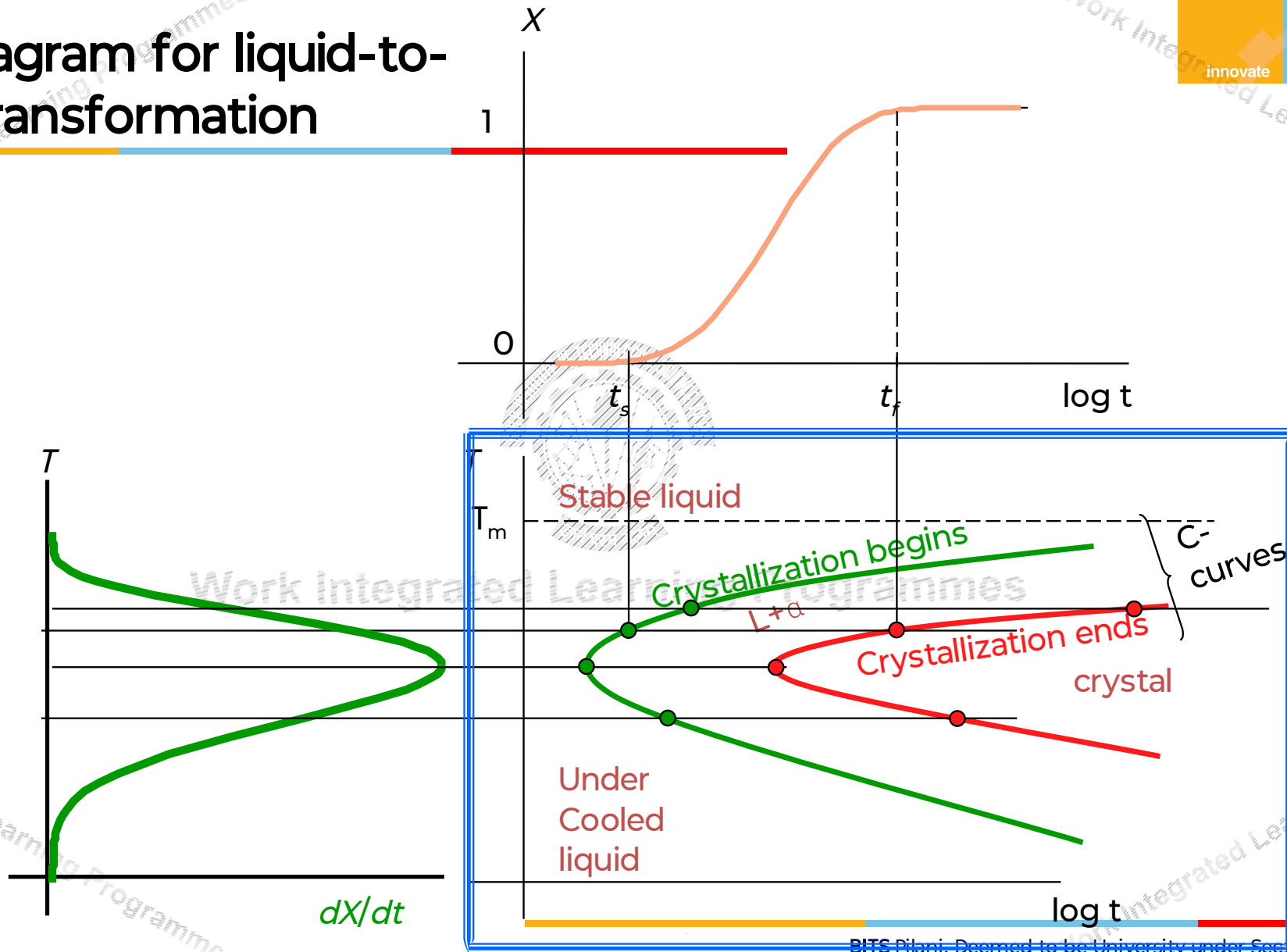
P Srinivasan
Core Engineering



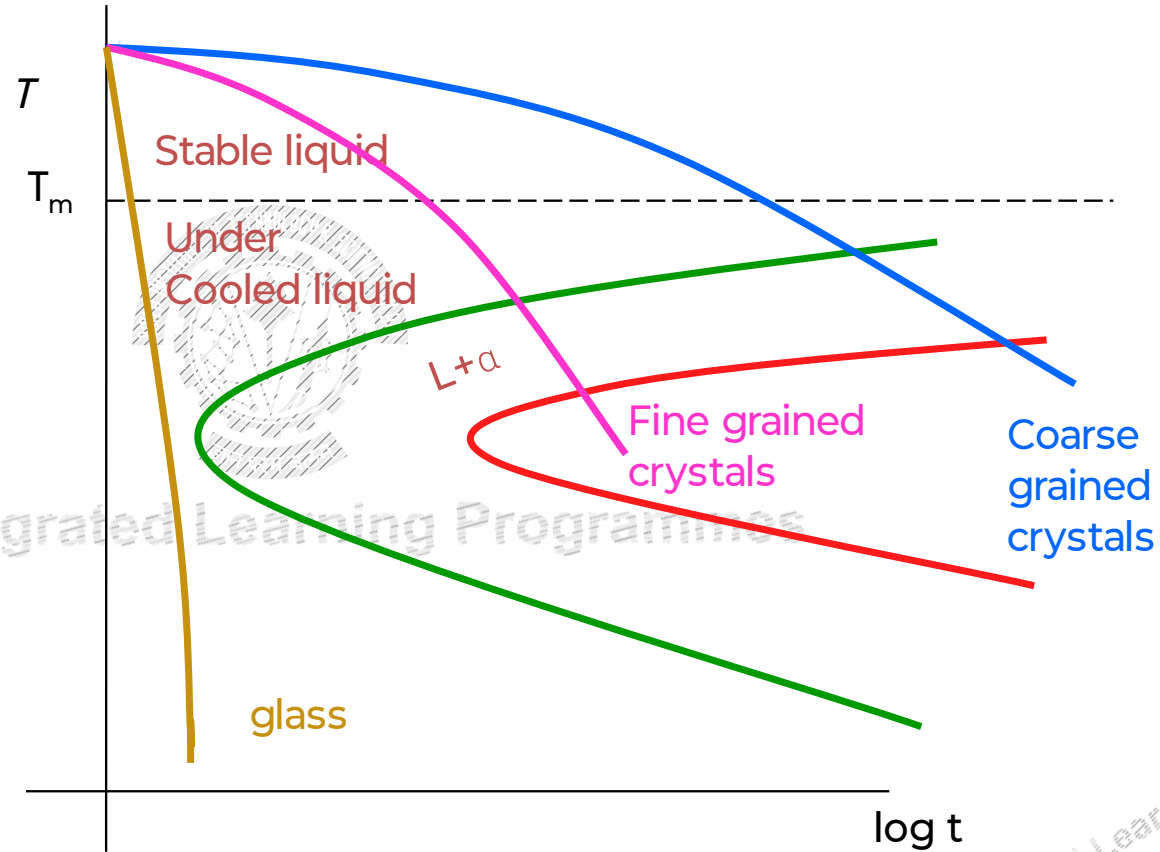
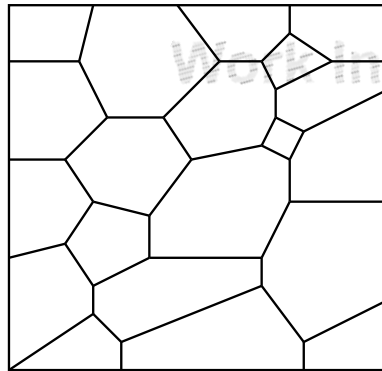
DEZG631 Material Technology & Testing

Lecture No. 2

TTT Diagram for liquid-to-solid transformation



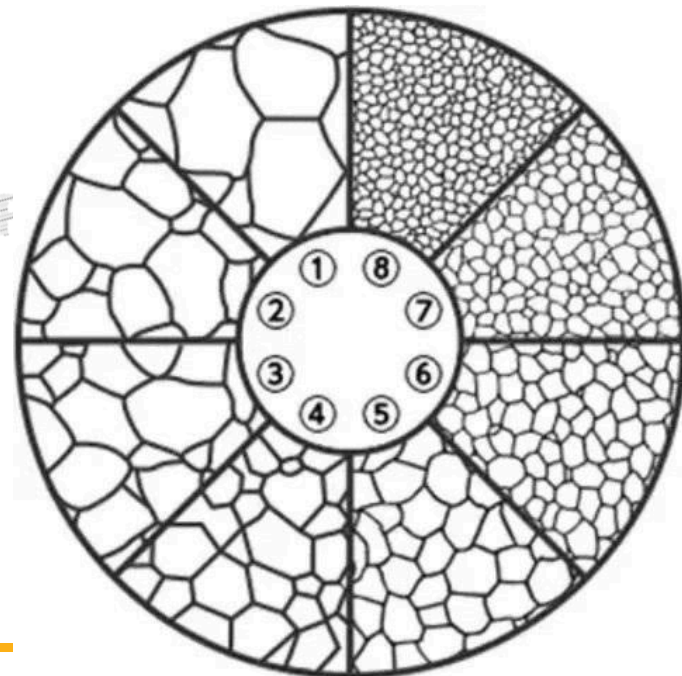
TTT Diagram for liquid-to-solid transformation



Grain structure....

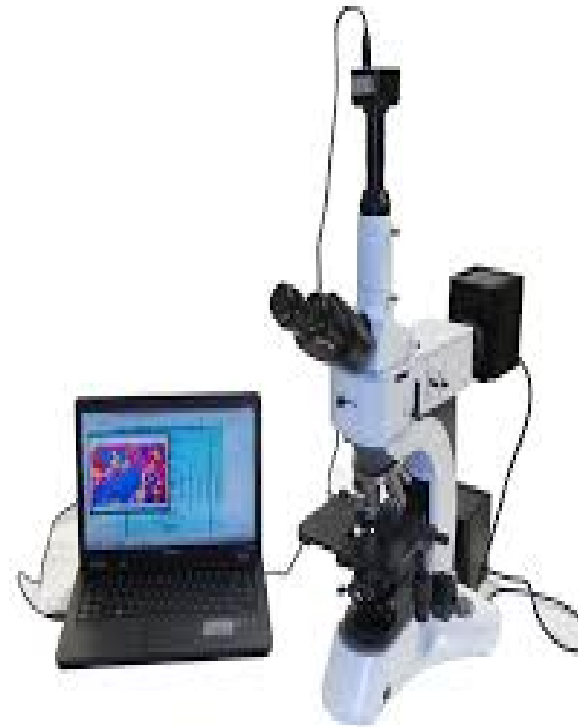
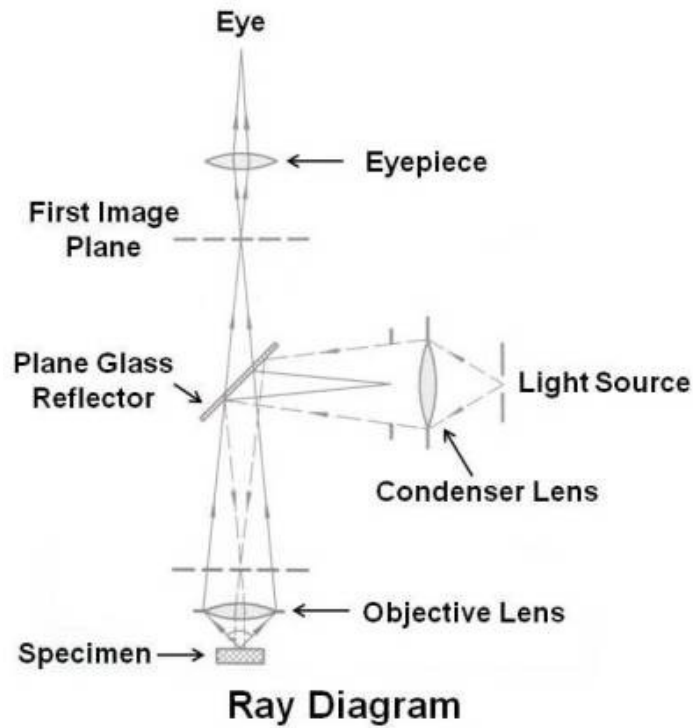
ASTM Standard E112

Grain-Size Number	Mean Number of Grains per Square Inch (at 100x Magnification)
1	1
3	4
6	32
10	512
12	2,048
14	8,192





Microscopic Examination



Microscopic Examination



- ❑ Microscopic examination is an extremely useful tool in the study and characterization of materials.
- ❑ Several important applications of microstructural examinations are as follows:
 - to ensure that the associations between the properties and structure (and defects) are properly understood
 - to predict the properties of materials once these relationships have been established
 - to design alloys with new property combinations
 - to determine whether or not a material has been correctly heat treated
 - to ascertain the mode of mechanical fracture

Optical Microscopy

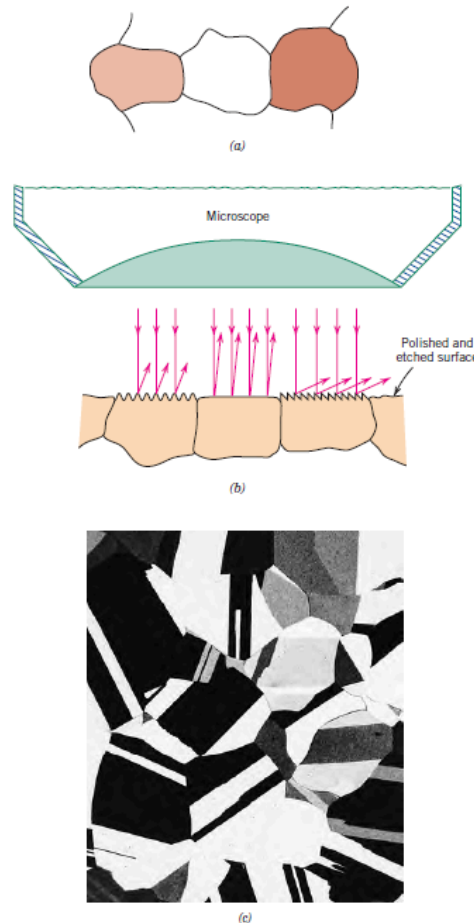


Figure 4.13

(a) Polished and etched grains as they might appear when viewed with an optical microscope. (b) Section taken through these grains showing how the etching characteristics and resulting surface texture vary from grain to grain because of differences in crystallographic orientation.

(c) Photomicrograph of a polycrystalline brass specimen. 60 $\times$. (Photomicrograph courtesy of J. E. Burke, General Electric Co.)

Optical Microscopy

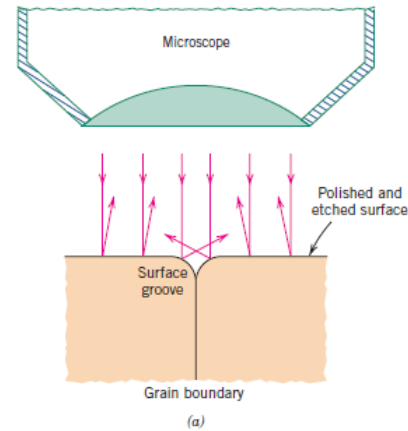
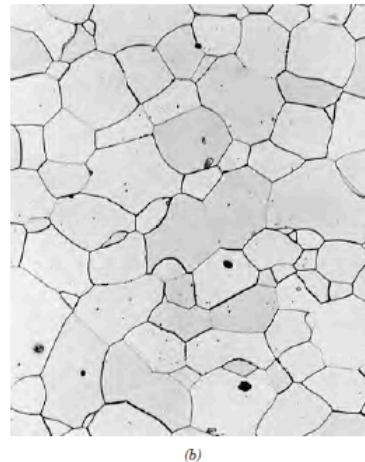


Figure 4.14 (a) Section of a grain boundary and its surface groove produced by etching; the light reflection characteristics in the vicinity of the groove are also shown. (b) Photomicrograph of the surface of a polished and etched polycrystalline specimen of an iron-chromium alloy in which the grain boundaries appear dark. 100 $\times$. [Photomicrograph courtesy of L. C. Smith and C. Brady, the National Bureau of Standards, Washington, DC (now the National Institute of Standards and Technology, Gaithersburg, MD.)]



Microstructure of different steel sheets:

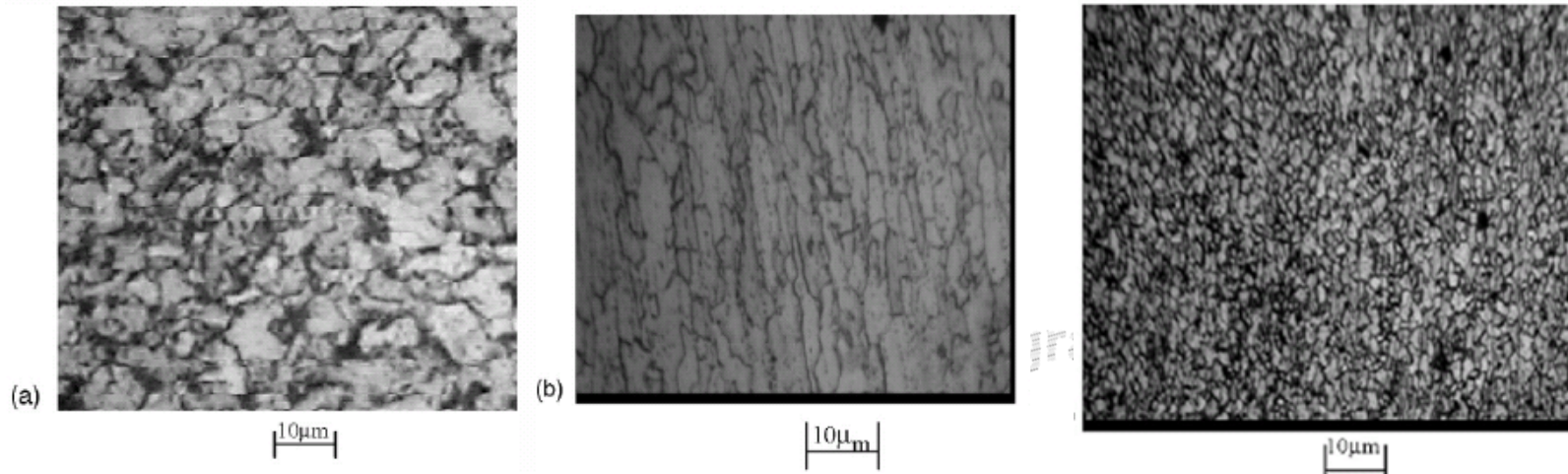
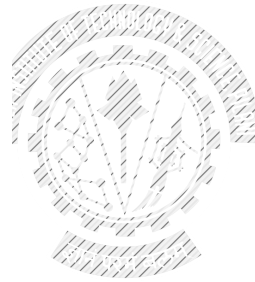
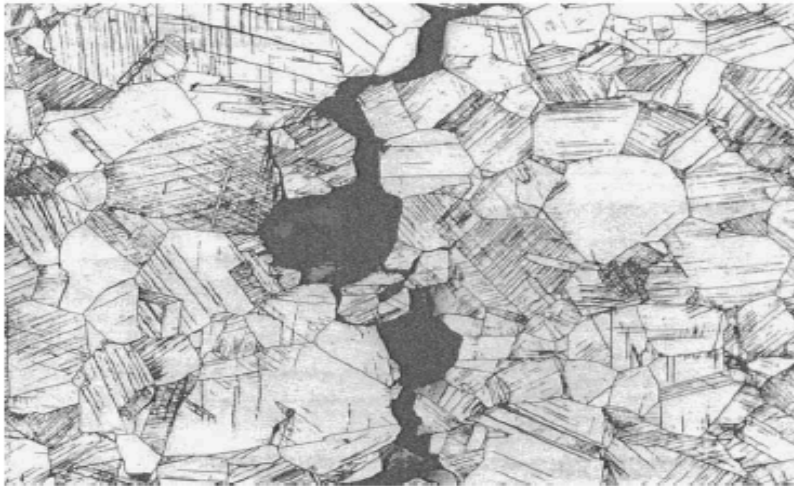
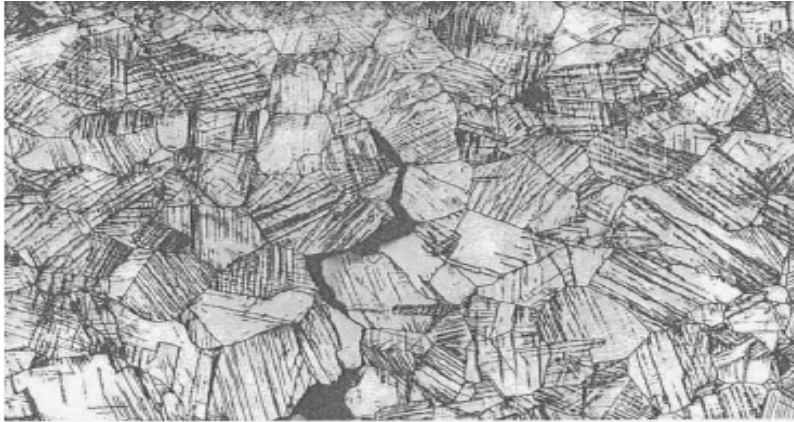


Fig: (a) microstructure of HSLA steel at magnification 400× nital etched
 (b) microstructure of carbon–manganese steel at magnification 500× nital etched
 (c) microstructure of micro alloyed steel at magnification 200× nital etched.

Microstructure of a crack



Work Integrated Learning Programmes

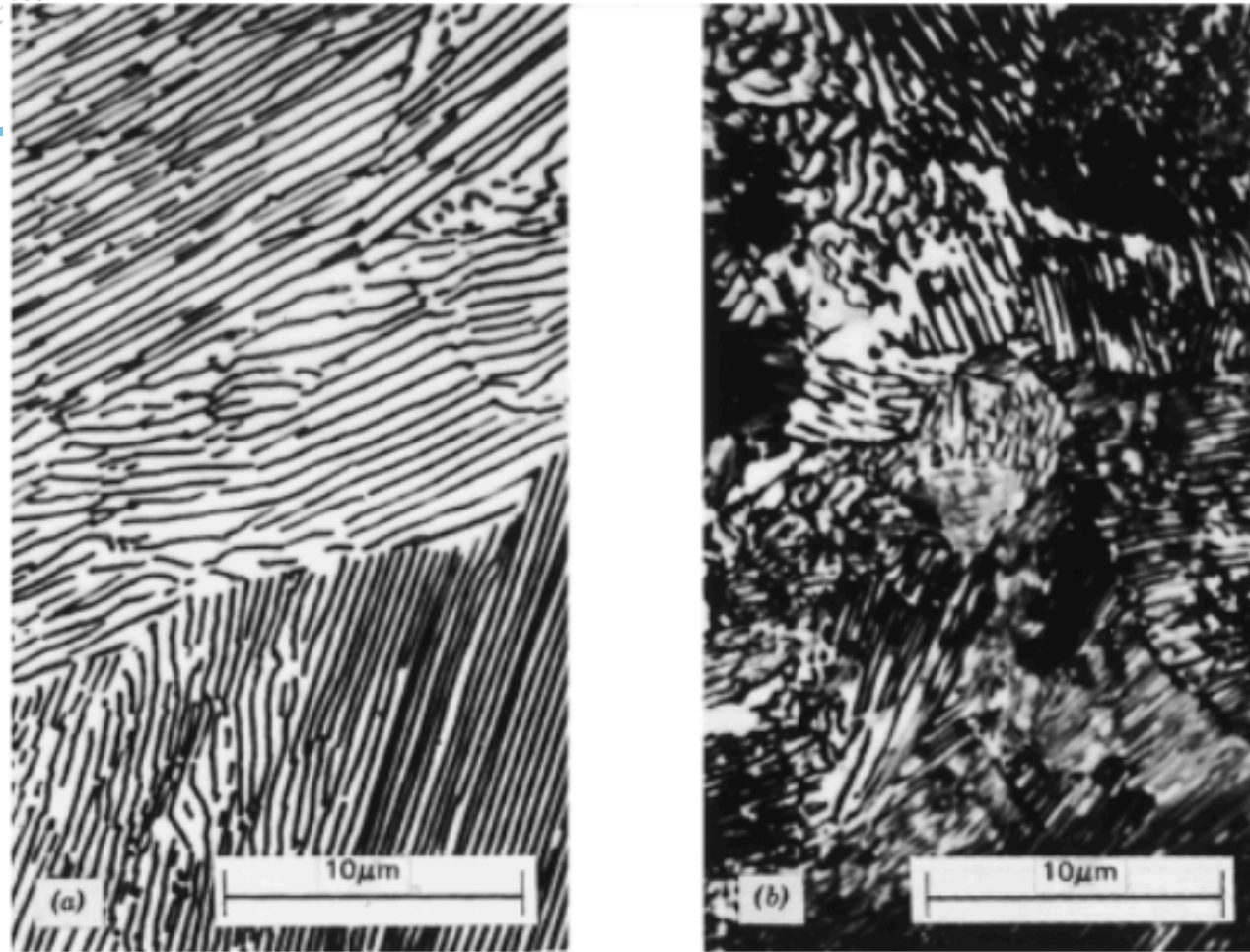


FIGURE 10.6 Photomicrographs of (a) coarse pearlite and (b) fine pearlite. 3000 $\times$. (From K. M. Ralls et al., *An Introduction to Materials Science and Engineering*, p. 361. Copyright © 1976 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.)

Callister

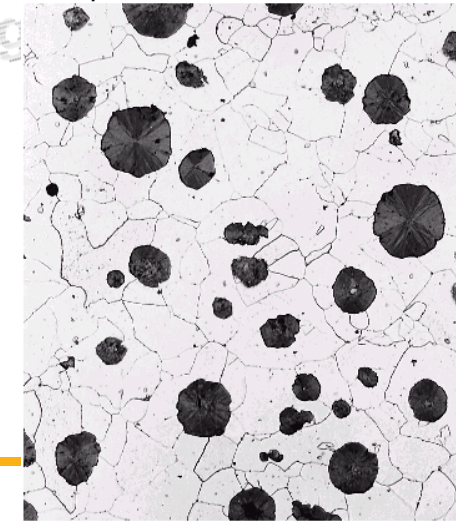
Types of Cast Iron

Gray iron

- graphite flakes
- weak & brittle under tension
- stronger under compression
- excellent vibrational dampening
- wear resistant



From Fig. 9.3(a) & (b), *Callister's MSE*,
Adapted Version.



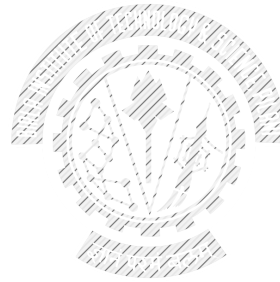
Ductile iron

- add Mg or Ce
- graphite in nodules not flakes
- matrix often pearlite - better ductility

Types of Cast Iron

White iron

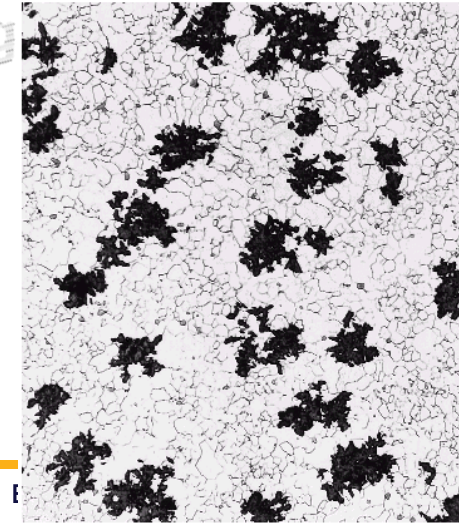
- <1wt% Si so harder but brittle
- more cementite



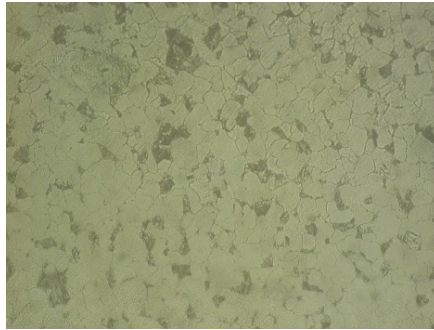
From Fig. 9.3(c) & (d), *Callister's MSE, Adapted Version.*

Malleable iron

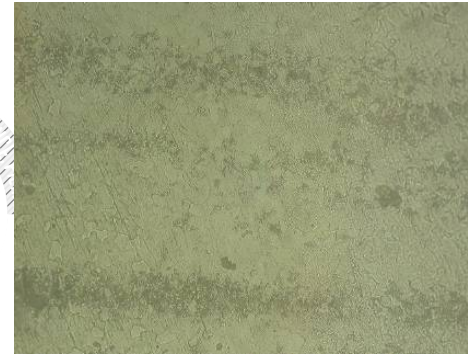
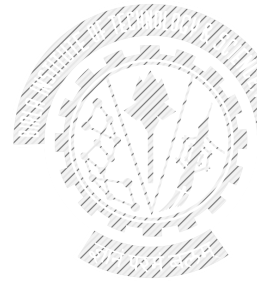
- heat treat at 800-900°C
- graphite in rosettes
- more ductile



Normal and deteriorated microstructure



normal microstructure (
good condition tube)
Which shows a typical
ferritic pearlitic structure



deteriorated
microstructure which
shows the
decomposition of the
pearlitic colonies

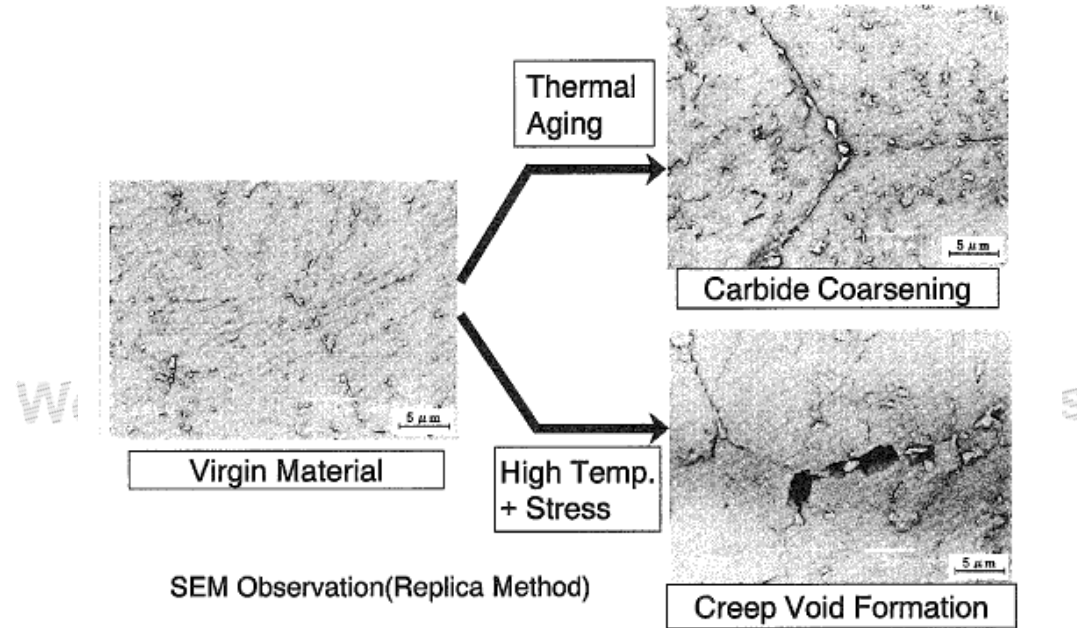


Figure 4: Example of microstructural changes caused by aged deterioration.

ASTM Grain Size Number



ASTM grain size number ' n ' is a measure of grain size.

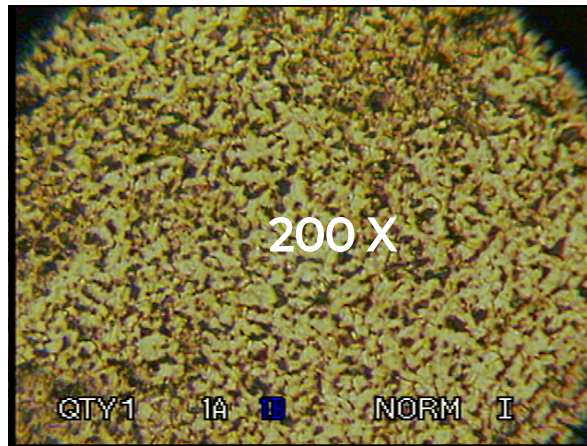
$$N = 2^{n-1} \text{ and } N' = N/15.5$$

N = Number of grains per square inch at 100X,

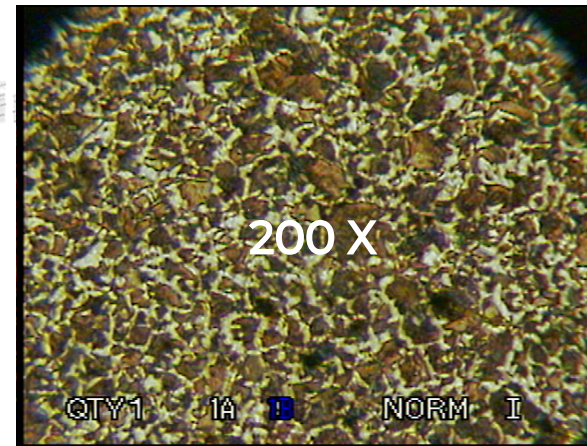
N' = Number of grains per square mm at 1X,

n = ASTM grain size number.

$n < 3$ – Coarse grained;
 $4 < n < 6$ – Medium grained
 $7 < n < 9$ – Fine grained;



1018 cold rolled steel, $n=10$



1045 cold rolled steel, $n=8$

ASTM Grain Size Number



ASTM grain size number ' n ' is a measure of grain size.

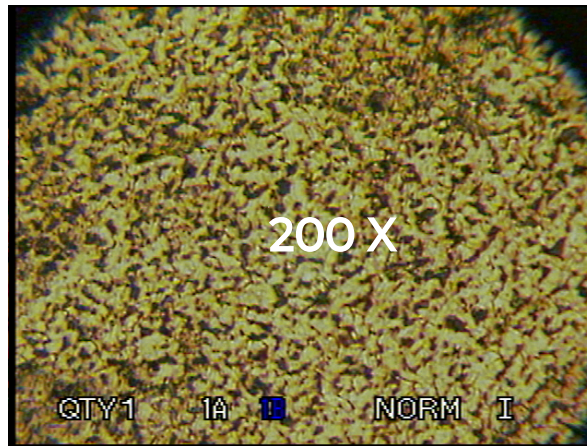
$$N = 2^{n-1} \text{ and } N' = N/15.5$$

N = Number of grains per square inch at 100X,

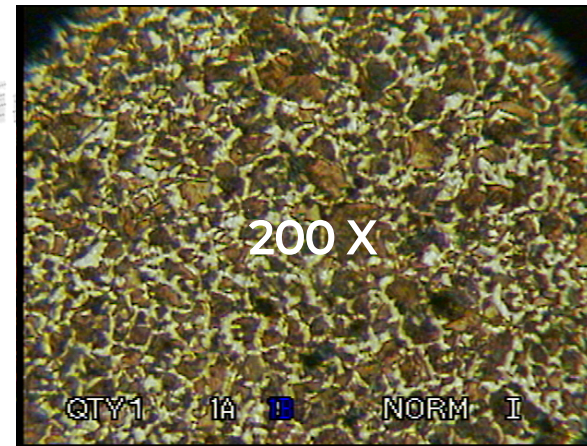
N' = Number of grains per square mm at 1X,

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 $7 < n < 9$ – Fine grained;



1018 cold rolled steel, $n=10$



1045 cold rolled steel, $n=8$

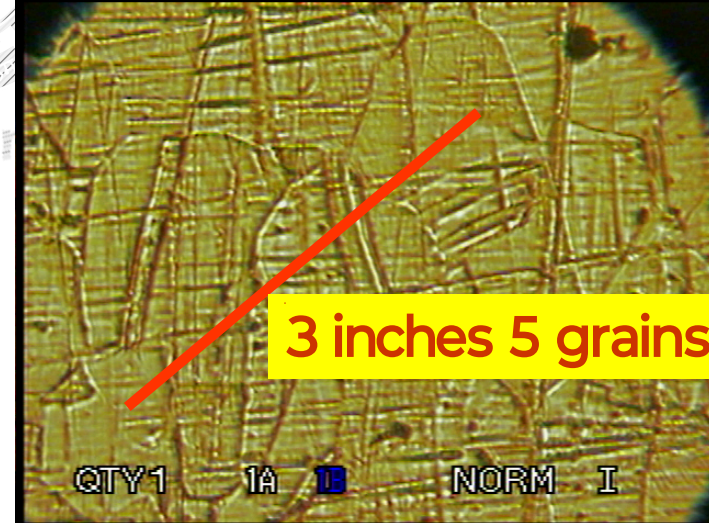
Average Grain Diameter



- ❑ Average grain diameter more directly represents grain size.
- ❑ Random line of known length is drawn on photomicrograph.
- ❑ Number of grains intersected is counted.
- ❑ Ratio of number of grains intersected to length of line, n_L is determined.

$$d = \frac{C}{n_L \times M}$$

$C = 1.5$, and M is magnification



Hall-Petch Equation



Relation between yield strength and grain size

Hall-Petch equation relates the grain size and yield strength of the material

$$\sigma_y = \sigma_0 + Kd^{-1/2}$$

σ_y = yield strength

d = average grain diameter of the grains

K & σ_0 are constants for the metal

The Hall-Petch equation is not valid for materials with very large or ultrafine grains

Exercise



If there 100 grains per square mm on a photomicrograph of a metal at 1X, what is the ASTM grain size number?

Solution

$$N = 2^{n-1}$$

Where, N = no. of grains per square inch at 100X

n = ASTM grain-size number

N is obtained by the following expression :

$N = N' / 15.5$ where N' = no. of grains per square mm at 1 X
given $N' = 100$ grains per square mm at 1 X

Thus , $N = 100 / 15.5 = 6.45$

$$6.45 = 2^{n-1}$$

$$\log 6.45 = (n-1) (\log 2)$$

$$n = 3.689$$

Exercise



If there 75 grains per square mm on a photomicrograph of a ceramic material at 2X, what is the ASTM grain size number of the material?

Solution $N = 2^{n-1}$

Where, N = no. of grains per square inch at 100X

n = ASTM grain-size number

N is obtained by the following expression :

$N = N' / 15.5$ where N' = no. of grains per square mm at 1 X

given $N' = 100$ grains per square mm at 1 X

If there are 75 grains per square mm at 2 X , then at 1 X we will have

$$N' = (2/1)^2 (75 \text{ grains/mm}^2) = 300$$

$$\text{Thus , } N = 300 / 15.5 = 19.3548$$

$$19.3548 = 2^{n-1}$$

$$\log 19.3548 = (n-1) (\log 2)$$

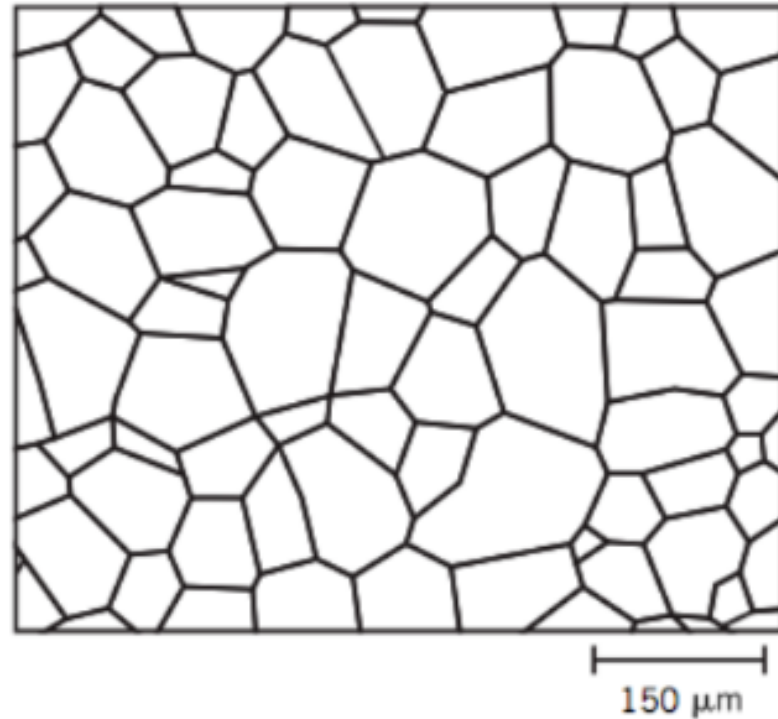
$$n = 5.2746$$

Exercise



Following is the schematic micrograph that represents the microstructure of some metal.

1. Using the intercept method, determine average grain size diameter. Use at least 5 straight line segments.
2. Calculate ASTM grain size number, n for this material



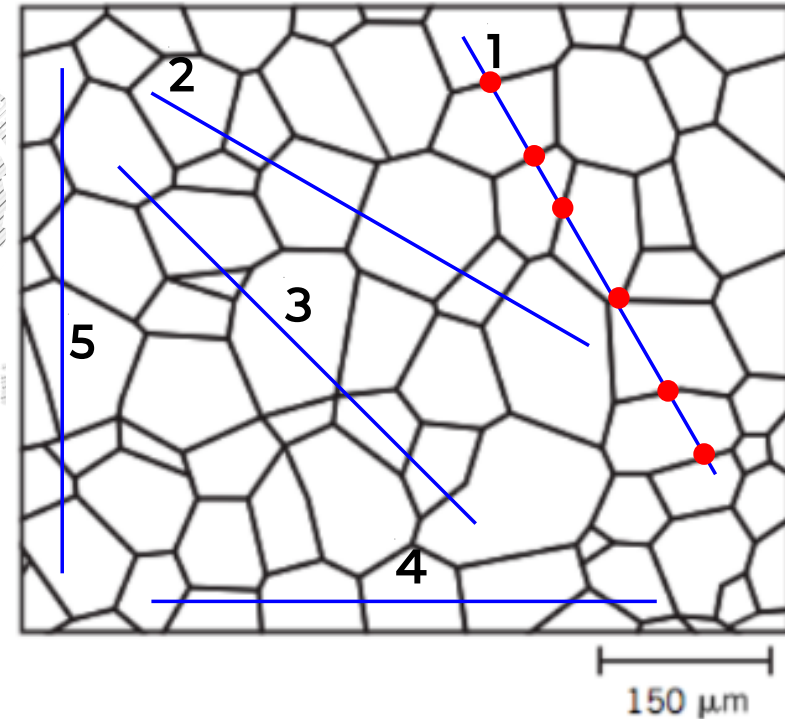
Solution



1. Using the intercept method, determine average grain size diameter. Use at least 5 straight line segments.

Step 1: Draw 5 lines of equal length

Step 2: Identify how many times each line has intersected a new grain



Solution



Step 3: Add up total number of intersections from all lines

Line	Number of Grain Intersected
1	6
2	5
3	7
4	6
5	7
Total	31